Module 1: Introduction of Automatic Sorting Pick & Place Station Based on Colour (ASPPSBC)

1.1. Introduction Brief Introduction of ASPPSBC Project

As the Mechatronics term is used which means intermixing of Mechanical and Electrical part. So, in our project our group is divided into different category to do a particular project. In which different members work on a particular task like PLC, Microcontroller, Designing, Panel work, Mechanical work.

During 6th semester we decided to make a project in which the main purpose of project is to sort and pick & place the different colour objects.

In this project a conveyor belt is used along with sensors which sense the objects moving on the conveyor belt and the robot hand at the end of belt picks and places the object at a specific location. In this a conveyor on which arm is to be mounted, which can pick and place the object and a pneumatic cylinder is used to sort one colour.

The project can be controlled with help of PLC and Microcontroller which gave particular signal to the device and run the system or components.

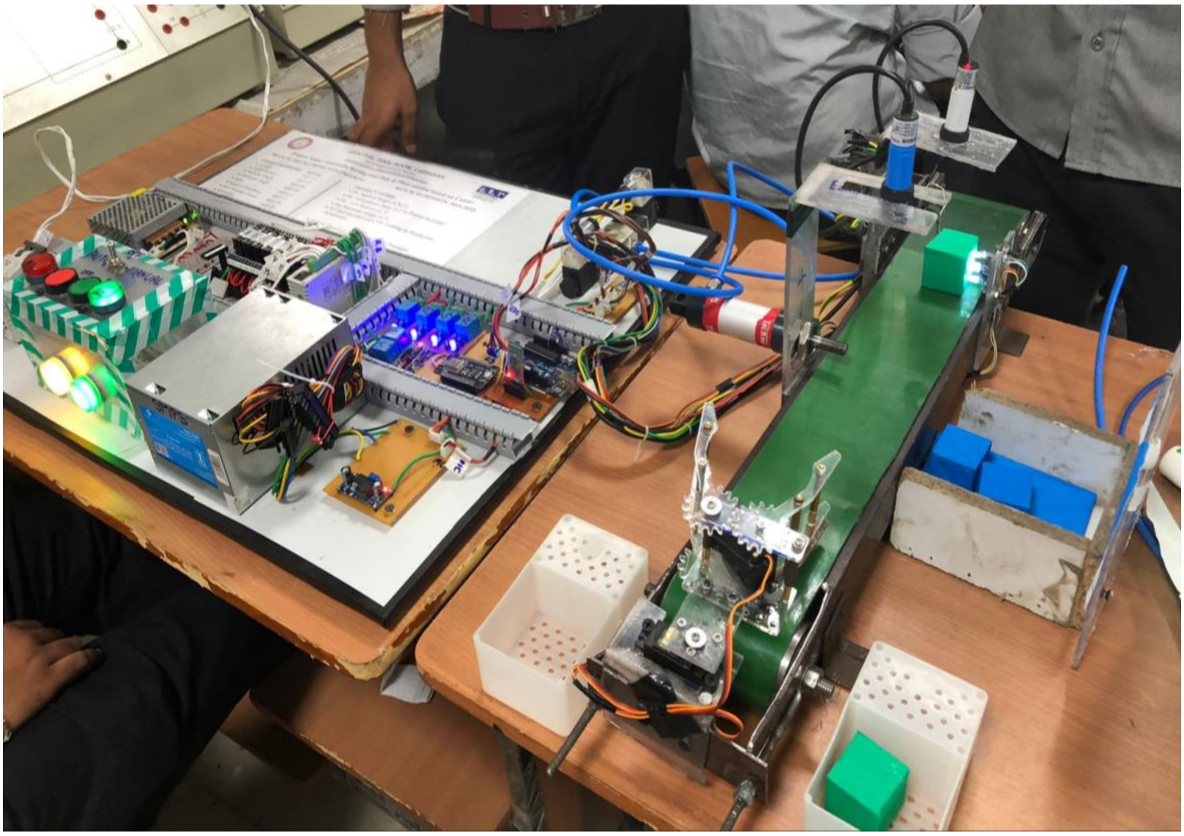

Figure 1.1.1 Project

1.2. Working of ASPPSBC Project

In this project we will learn the process of Sorting, Pick and Place using a robot end effector using a conveyor belt. On conveyor belt colour sensor is present which will sense the object whether it is to be passed or it is to be sorted. When the object is passed through colour sensor it comes across a pneumatic cylinder and a proximity sensor. Here the sensor will sense the object whether it is required or not, if the object is unwanted the pneumatic cylinder will sort that particular object and hence will allow the needed object to pass through it.

The main working of project is defined as follow:

- First, we can turn on supply with help of MCB.
- After that the system is on which can start with help of two modes Manual and Auto mode. The mode can switch with help of toggle switch. In auto mode the signal is given with help of Mobile app.
- Now we select manual mode. After green push is pressed the conveyor is run and setup is start.
- Now we can place colour object on conveyor after that is object can be detected by sensorand scan the colour of object. After scanning done the conveyer start.
- If colour is red or green then the object is passes to the gripper and then the gripper picks and place the object on particular station, if the colour of object is blue then second sensor sense the object and turn on the pneumatic cylinder which can sort the blue object.
- To stop the whole cycle or project we can press red push button and turn off the MCB.

This is the main working of our project which can be controlled with help of PLC and Microcontroller.

1.3 Applications of ASPPSBC Project in Industry

As we see in our daily technology is increase day by day. The main objective of technology is to do more work in a short time. Thus, we can make a project of colour sorting and pick and place station in which a conveyor is used for the movement of object. All the setup is run with help of PLC and Microcontroller.

There is various application of project which can be used in industry as defined below:

- **STORING:** A simple pick and place task is to program your robot to store parts that come off a production line or from a previous operation. This is great example of a non-value-added task and is therefore perfect for a robot.

- **SORTING:** A great pick and place task is to sort different objects into separate piles or onto separate conveyors. This usually requires robot vision to detect the various object types. However, this is not always the case; Sorting can also be combined with other tasks such as packaging, to allow the robot to sort parts as it moves them to the next stage in their operation.

- **PACKAGING:** Packaging consists of picking objects from one part of the workspace and placing them into a box, tray, bag, or another packaging container. It is probably one of the most common pick and place tasks.

- **QUALITY CONTROL:** Pick and place is often used in quality control tasks immediately following an inspection step. The inspection system identifies which products are defected and the robot moves them into a location for rework or rejection.

- **PALLETIZING:** Palletizing is a very specific type of pick and place task that involves arranging packaged products or other objects onto a pallet or tray. The key difference with palletizing is the structured pattern in which the parts are arranged. Palletizing programming tools make it easy to create the required pattern or pallet.

- **DE-PALLETIZING:** A similar task to palletizing, de-palletizing simply means taking items off a pallet and placing them in a way that is useful for the next stage in the process.

Module 2: Mechanical Design and Manufacturing

2.1 Bill of Material

The material used in mechanical category major project is listed below in table 2.1.1

S. no.	Material	Specification	Unit	Qty
1.	BELT	Green PVC belt, 5mm thickness	Meter	4
2.	CHAIN TIGHTER	Steel piece	Nos.	02
3.	ROLLER	Metallic steel	Nos.	02
4.	MOTOR	DC Gear Motor 12V DC Operated	No.	01
5.	PVC STANDS	Hard plastic	Meter	1
6.	BEARING	Stainless steel bearing	Nos	04
7.	ALUMINIUM CHANNEL	Aluminum channel 5mm thickness	Sq. Meter	1*1
8.	NUT BOLT		Nos.	10
9.	WARSHEL	Steel iron	Nos.	20
10.	KNURLING TOOL		No.	01
11.	SCREW	Gypsum screw	No.	01
12.	NUT BOLT	Allen nut bolt 2mm	Meter	1

Table 2.1.1

2.2 Introduction to SolidWorks

- **Introduction to Software.**

  SolidWorks is a solid modelling computer-aided design and computer-aided engineering application published by Dassault Systems. The software is used for planning, visual ideation, modelling, feasibility assessment, prototyping, and project management. The software is then used for design and building of mechanical, electrical, and software elements. Basically, we have used SolidWorks 2018 as offers Rules-Based Machining and Automatic Feature Recognition that result in easy-to-learn, yet fast and powerful CAM**.**

Figure 2.2.1 Software logo

- **Benefits of SolidWorks: -**

- Increase Productivity. SOLIDWORKS 3D modelling allows you to simulate the design aspect in 3D right away and make any necessary adjustments right away.
- Design Better.
- Cost-Efficient.
- Enhanced Collaboration.
- Innovative.

- **What is Design in Engineering?**

  Design in mechanical engineering refers to creating, designing and building new machines that improve the efficiency of existing ones. Some of the biggest human achievements, from smart cars to the international space station, were possible because of the innovative design and unique thinking of the world's top mechanical engineers. As a mechanical engineer working in design, you can create solutions to solve a specific problem or address a particular human need.

- **Mechanical Design According to Project**

  Mechanical design and drafting courses are basic courses for engineering majors. According to different major, emphases of mechanical design and drawing courses are different. However, the main purpose is to cultivate spatial imagination ability, to guide students’ solid modelling and 3D construction and to cultivate engineering and technical personnel for the enterprises, who possess modern design philosophy and creative thinking ability. The course, with a strong theoretical and practical quality is aimed at developing students’ ability to think, analyses and raise questions about engineering as well as improving the capacity to draw and read engineering chartings through theoretical study and practical experience.

their groundwork for further professional curriculum study and future career practice well done.

We got the idea of our mechanical design by searching online which material are available and which design would be best suitable for our project with accurate dimensions and this was possible to know with the help of 3D Design Software SolidWorks 2018. Different parts of the project were designed separately and then assembled in the software itself. where the accurate mechanical design was formed and later on, we were able to give our project the final size shape and dimensions.

- **Need of Mechanical Design.**

Mechanical design services provide the necessary components and framework to achieve the product’s intended functions. In addition, manufacturers will oftenreject these files, which can create costly redesign and engineering down the road. All of this can be minimized from the start, by utilizing a professional design and engaging their mechanical design services. Durability is defined by different materials, fasteners, assembly components, etc. Function is achieved through gears, circuit boards, and other mechanisms at play within the invention. An experienced mechanical designer can identify points of breakage in an industrial design, and plan out the necessary internal workings to ensure the product will last, while effectively performing its functions. Thus,it is more effective to combine mechanical and industrial design simultaneously to develop an aesthetic, durable and functioning prototype.

- **Mechanical Parts Design.**

In mechanical part we can design of conveyor for our project which can be defined as follow in the figures:

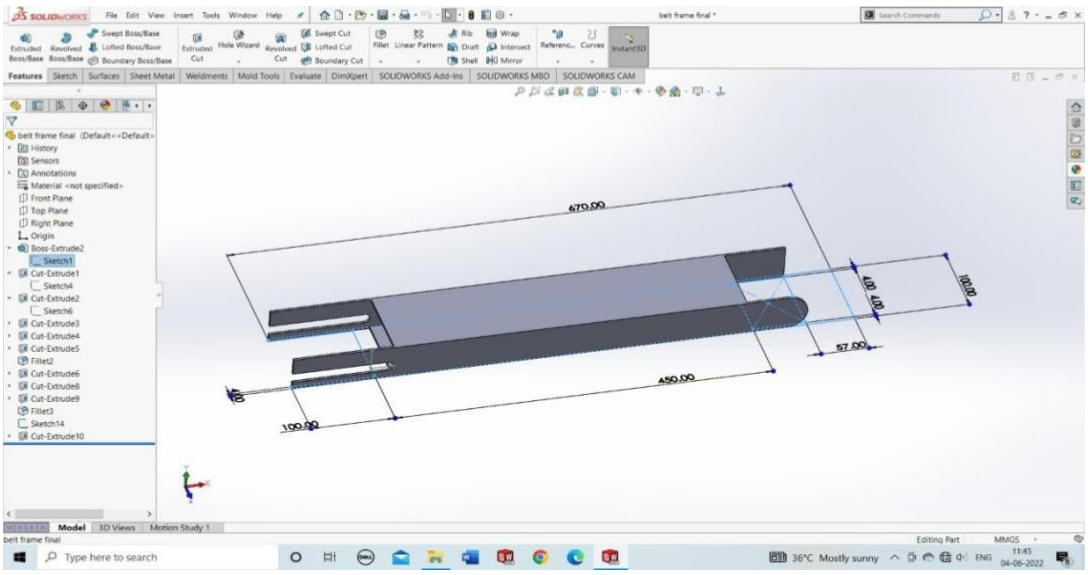

Figure 2.2.2 Frame

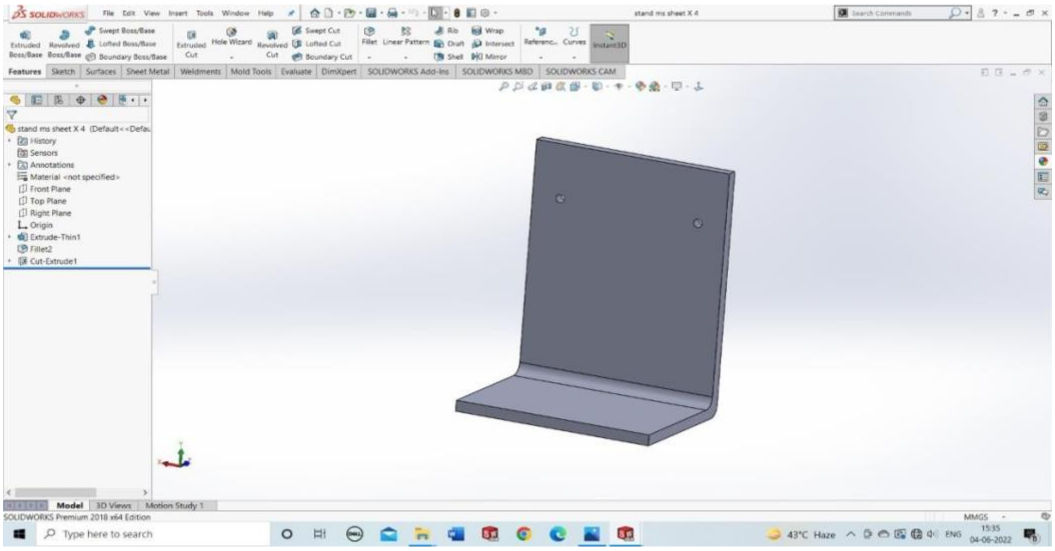

Figure 2.2.3 Stand

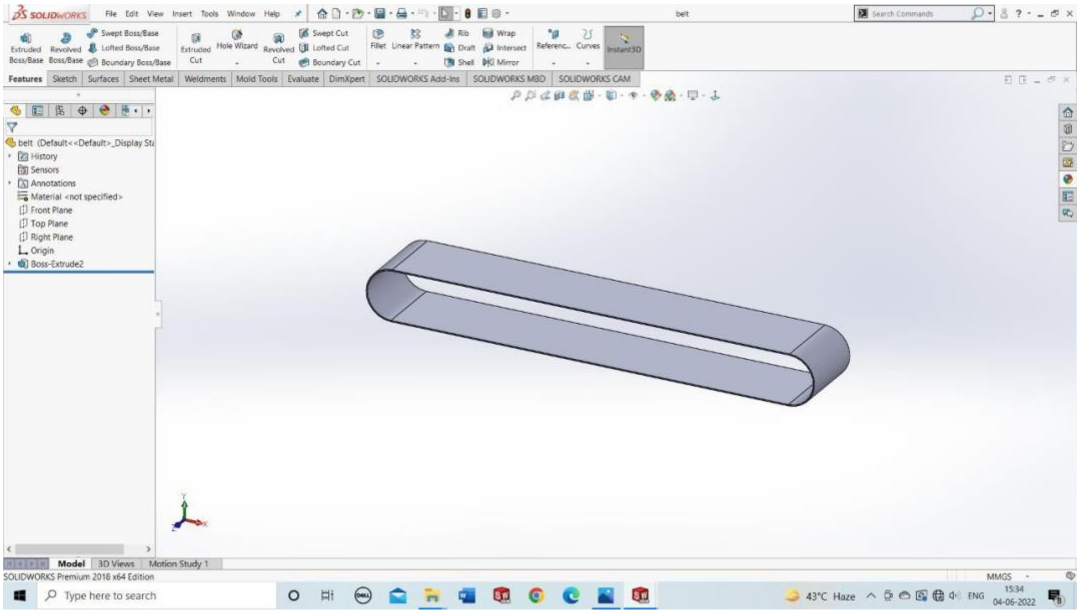

Figure 2.2.4 Belt

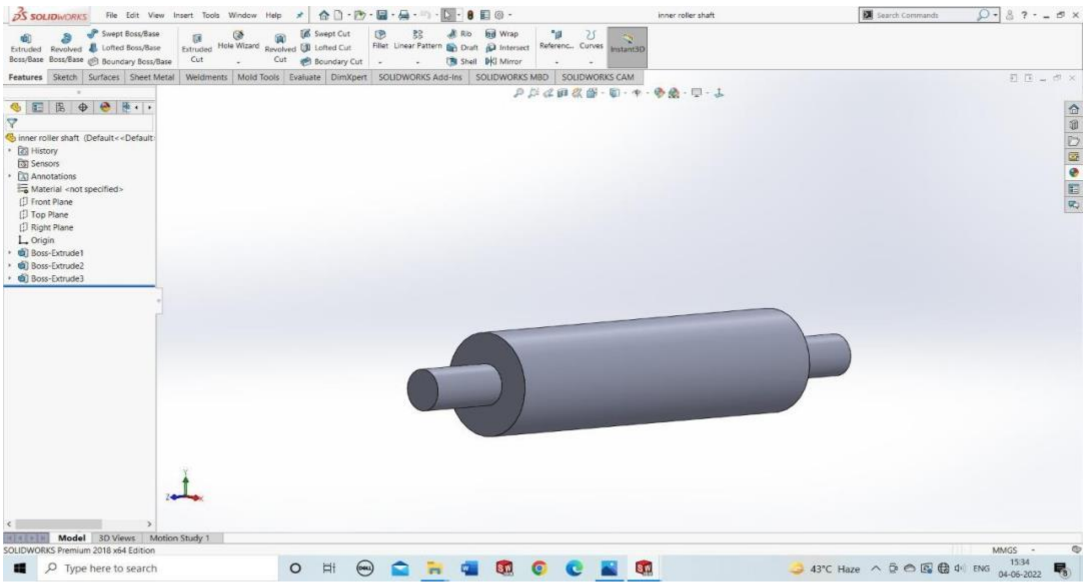

Figure 2.2.5 Roller

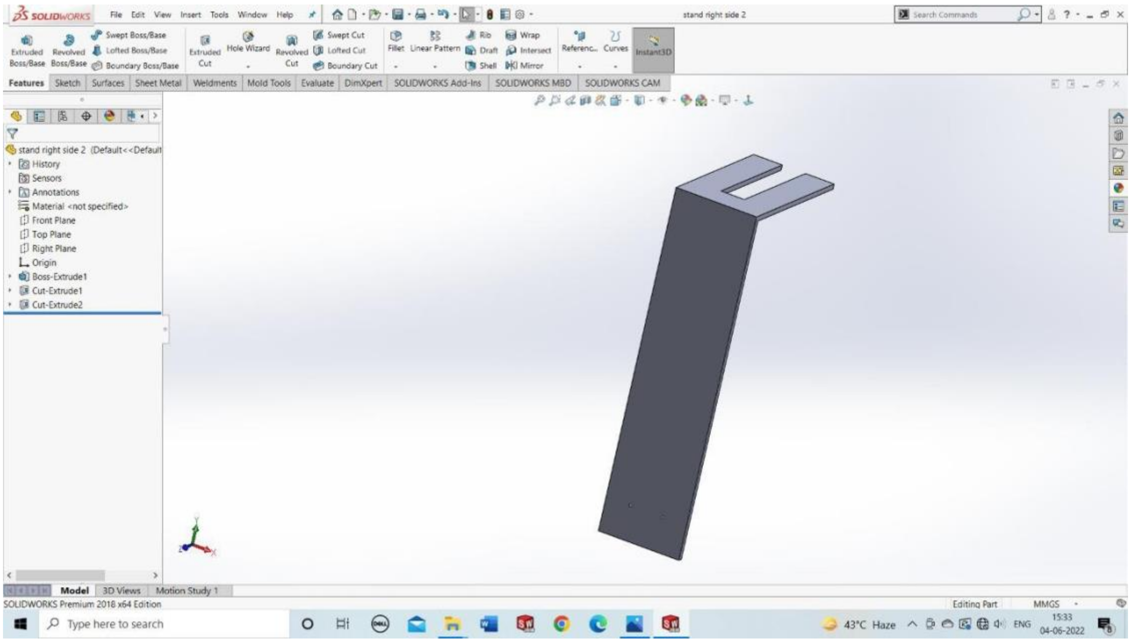

Figure 2.2.6 Sensor stand

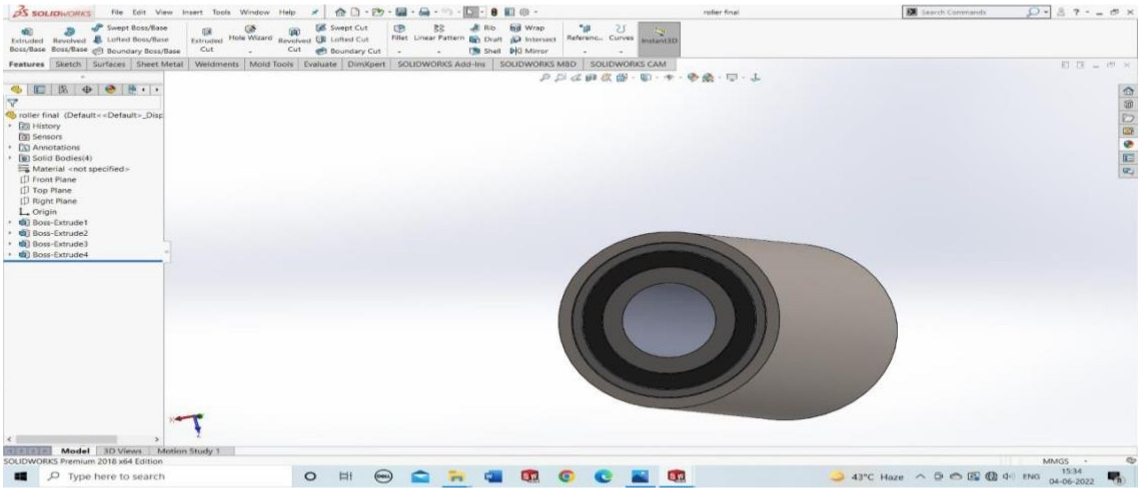

Figure 2.2.7 Roller

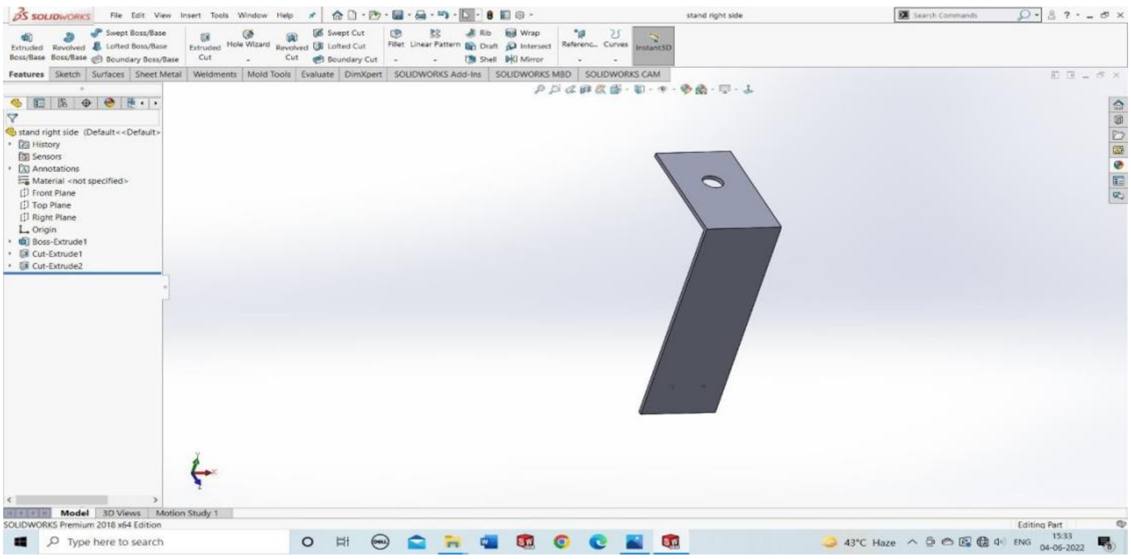

Figure 2.2.8 Sensor Stand

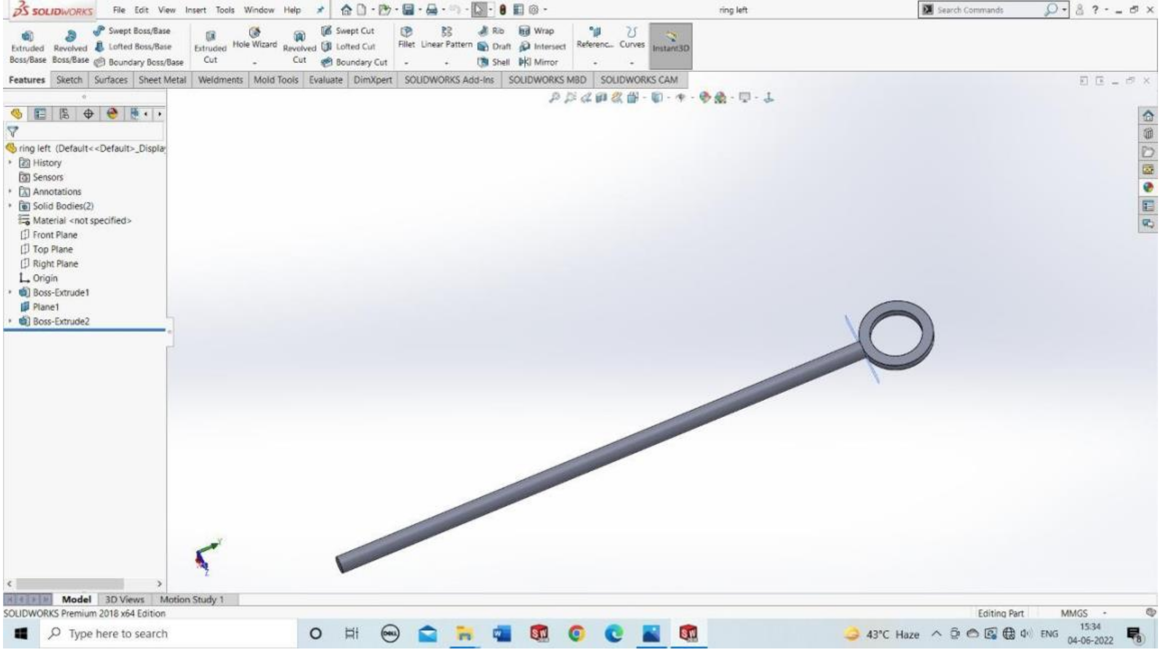

Figure 2.2.9 Adjustment screw

2.3 Different Manufacturing Process Used in ASPPSBC Project

There are various processes which can we used in our major project to make conveyor.

- **Knurling**

  It is a process used to create any combination of horizontal, vertical lines on the surface of work piece. The metal modification process of creating small ridges along a metal object's surface to improve the ease with which such an object can be physically handled, lifted or gripped.

- **Drilling**

  It is a material removing or cutting process in which the tool uses a drill bit to cut hole of circular cross section in solid materials. The drill is usually a rotating cutting tool, often multi-point. The bit is pressed against the work piece and rotated at speeds of hundreds to thousands of revolutions per minute. As a result, the cutting edge is pressed against the work piece, which removes chips from the hole during drilling.

- **Turning**

  Turning is the most common lathe machining operation. During the turningprocess, a cutting tool removes material from the outer diameter of a rotating work piece. The main objective of turning is to reduce the work piece diameter to the desired dimension.

- **Facing**

  Facing is the process of removing material from the end and/or shoulder of a work piece, using a special tool to produce a smooth surface perpendicular to the rotational axis of the work piece.

- **Boring**

  Enlarging an existing hole by subtracting material from the inner diameter. Typically, not used for holes smaller than a few centimeters in diameter but can achieve tight tolerances, similar to the combination of drilling and reaming.

- **Threading**

  Creating threads in a hole into which a screw will be placed. Required when screws are used in metal part. Hole size, pitch, and type of thread must be known (it is recommended to pick a standard screw size). Creating the threaded hole is known as tapping.

- **Bearing fitting**

  Bearing reduce friction and make rotation smoother. It is installed in roller and shaft.

- **Belt fitting**

  Belt is to transfer power from one source to another. The belt is installed on rollers to move the boxes.

- **Stand fitting**

  Stands are installed at the side of aluminum sheets to mount different parts like sensors, gripper.

- **Making of shaft according to bear size**

  Shaft is made for proper rotating.

- **Mounting of motor**

  Motor is mounted by Bosch screw fitting on aluminum sheet.

Module 3: Electrical Panel Wiring & PLC Programming

3.1 Bill of Material

The material used in PLC category major project is listed below in table 3.1.1

Table 3.1.1

S. no.	Material	Specification	Unit	Qty
1.	PLC	DELTA MODEL=DVP-14SS2 TRANSISTOR TYPE	No.	01
2.	COMMUNICATION CABLE	RS 232 to din 8 and USB to male din 9 pin	No.	01+01
3.	INDICATION LED	Yellow, Green 230V AC Red, Green 24V DC	No.	01+01 01+01
4.	TOGGLE SWITCH	SPDT 240V AC 20amp	No.	01
5.	PUSH BUTTON	Red and Green with NO and NC	No.	01+01
6.	SMPS	24V DC – 2 amp and 12V DC - 15 amp + 5V DC - 30 amp	No.	01 01
7.	MCB	DP	No.	01
8.	RELAY CARD	8 Channel Relay card module	No.	01
9.	SENSOR	Proximity sensor 24VDC PNP (NO, NC) Proximity sensor 24VDC NPN (NO, NC)	No.	01 01
10.	DCV	Techno = 5/2-way Double Solenoid valve 24V DC operated	No.	01
11.	PNEUMATIC ACTUATOR	Techno = Double acting Pneumatic Cylinder 25*25, 40mm stroke	No.	01
12.	DIN RAIL	Steel din rail	Meter	1
13.	MOTOR	DC gear motor 12V DC operated	No.	01
14.	WIRE	Copper pvc insulated wires Diameter (1mm, 0.75mm, 0.5mm)	Meter	12
15.	CONNECTOR	Plastic 8-way input and output Male and female wire connector	No.	01 02
16.	BUCK CONVERTOR	12V Dc to 0 to 12 V DC	No.	01

3.2 Circuit Diagram of Electrical Panel

First, we can use Auto cad electrical for make the panel design. After the completing the design further electrical panel or connection of component will make. There are two designs whichcan be made in Auto cad as defined in figure:

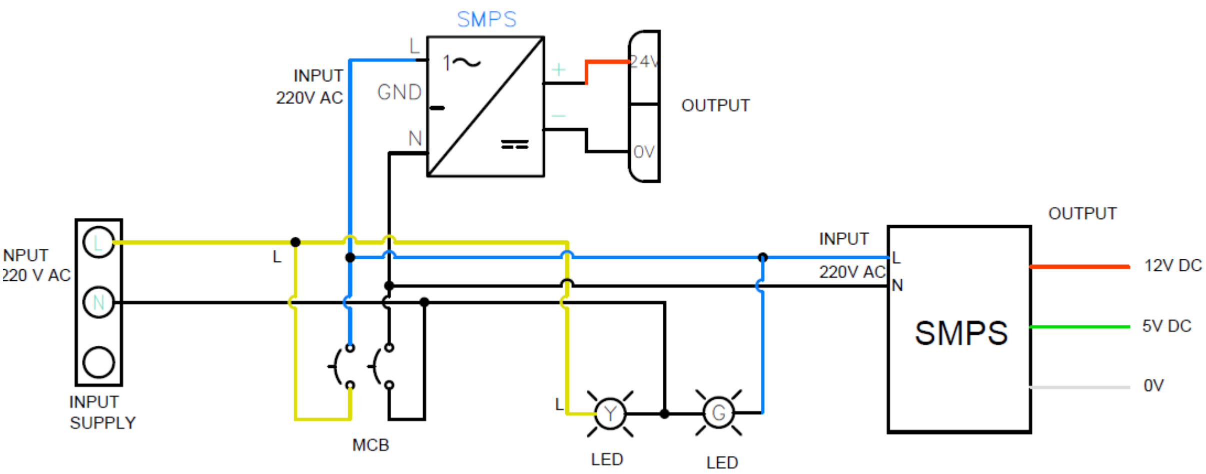

Figure 3.2.1

In figure 3.2.1 main supply passes through the connector. MCB use to control the flow of supply. When MCB is on it can turn on the both smps which can be used in panel and alsoglow the green led.

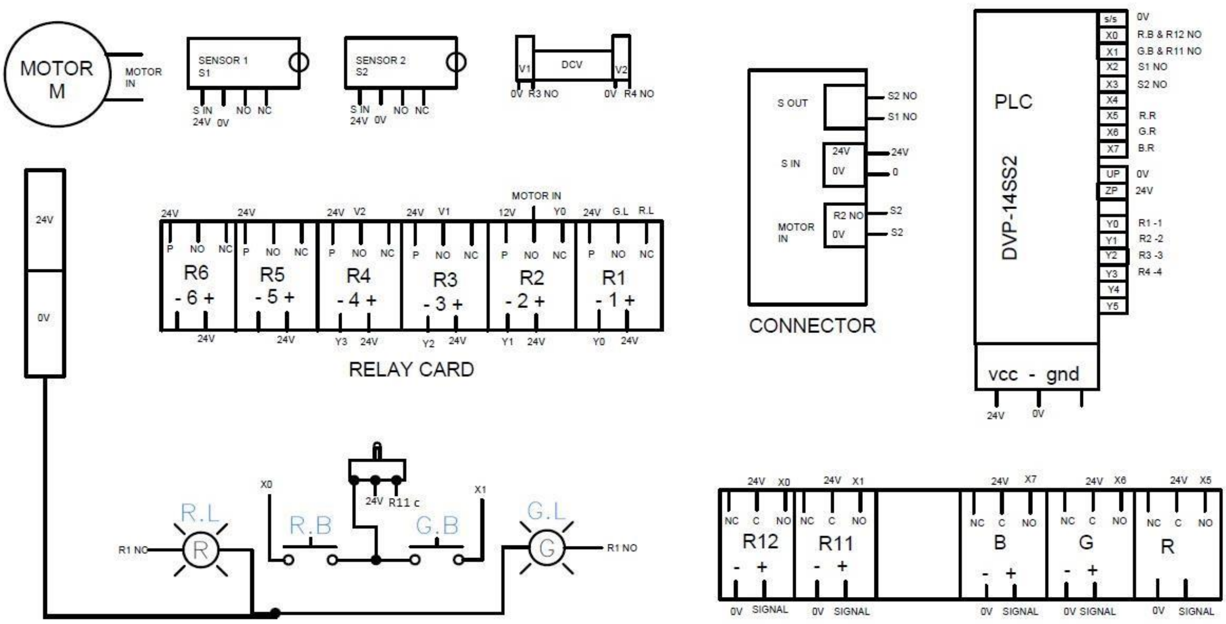

Figure 3.2.2

In the figure 3.2.2 the electrical panel diagram is design which can show about the connection of components.

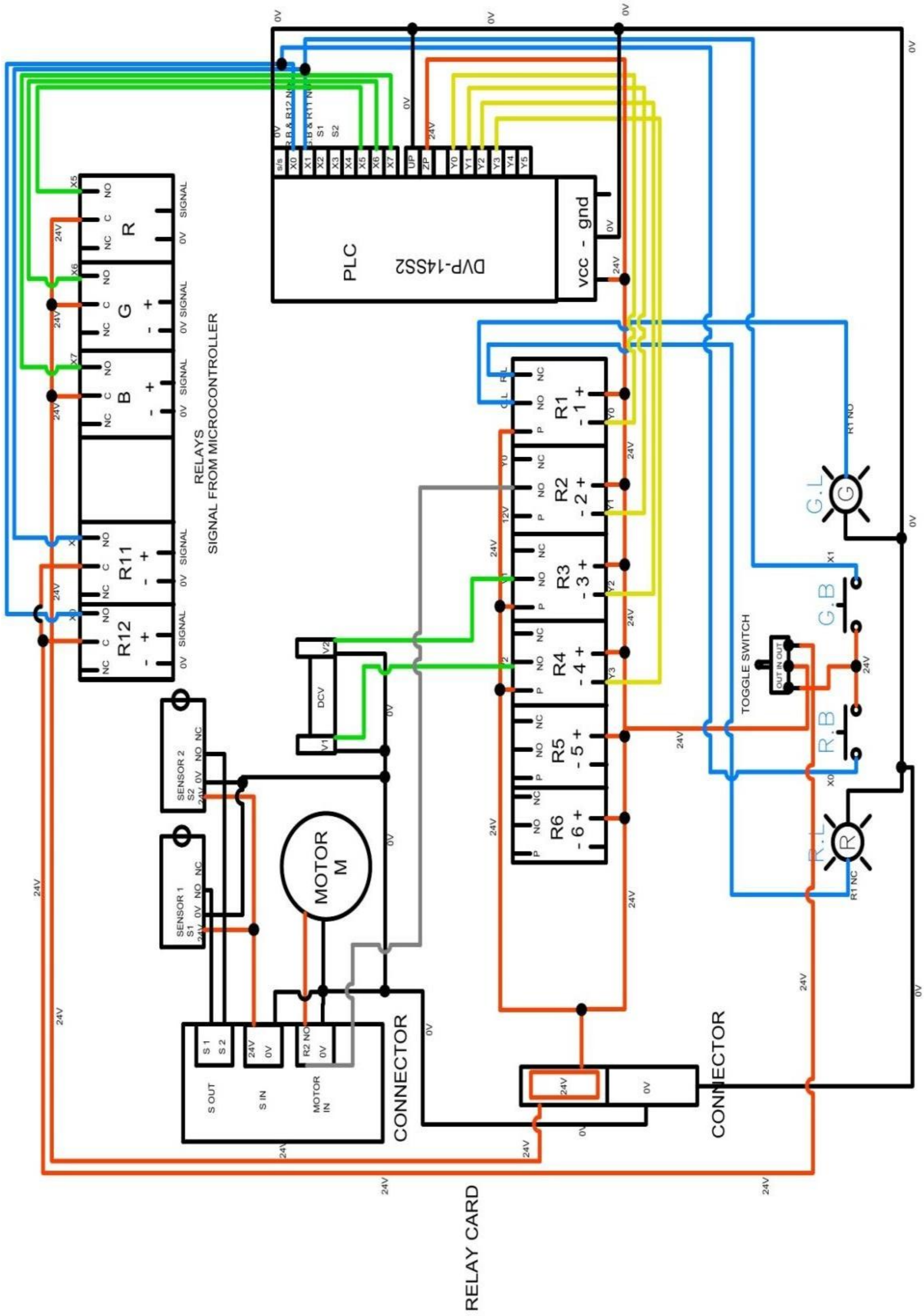

Figure 3.2.3 Panel wiring

3.3 Introduction to Different Electrical & Pneumatic Components

In our major project we can use different electrical & pneumatic components which can be defined in BOM.

3.3.1 Pneumatic Components

We can use some pneumatic components acc. to our project work requirement. The main function of pneumatic components is to sort an object or material. The components are defined as follow.

- **Pneumatic Actuator**

  We can use double acting cylinder for sorting material in the project.

  A double-acting pneumatic cylinder has two ports that control the movement of the rod. The compressed air causes the rod to move in two directions by extending and retracting without the assistance of a spring.

Figure 3.3.1 Pneumatic cylinder

Figure 3.3.2 DCV

- **5/2 Double Solenoid DCV**

  The main objective of DCV is to control the pneumatic cylinder used in project.

  5/2-way valves are used to actuate double acting pneumatic actuators, such as pneumatic cylinders, rod less cylinders, grippers and rotary actuators. Double acting actuators require compressed air to move in both directions.

- **Pneumatic Pipe**

  Pneumatic pipe play an important role in working. It helps to transfer the compressed air from one place to other.

Figure 3.3.3 Pipe

Figure 3.3.4 Compressor

- **Pneumatic Compressor**

  Pneumatic compressor plays an important role where pneumatic application is used. It helps to compress the environment air and store in it for further use.

3.3.2 Electrical components

In the project there are various electrical components which we can used acc. to projectworking conditions and requirement. The electrical components which we can use are detailed as follow:

- **PLC**

  A programmable logic controller (PLC) is an industrial computer control system that continuously monitors the state of input devices and makes decisions based upon a custom program to control the state of output devices. In the project we can use DELTA PLC Model DVP-14SS2. It can control the panel and conveyor signals.

Figure 3.3.5 PLC

Figure 3.3.6 Push button

- **Push Buttons**

  Push buttons can be explained as simple power controlling switches of a machine or appliance. With use of push button, we can switch on and off the conveyer and their controls. We can use red and green push button for start and stop.

- **Toggle Switch.**

  A Single Pole Double Throw (SPDT) switch is a switch that only has a single input and can connect to and switch between the 2 outputs. Toggle switch is use to control the signal for manual and auto mode.

Figure 3.3.7 Toggle switch

Figure 3.3.8 Led

- **Indication LED's**

The LED indicator is located at your device. It can display various colour such as blue, red and green. In project there are different led used which can convey some message according to work. In panel yellow led indicate that main supply is coming, green led is on after MCB is on, red led indicate that PLC system is off and green led show that system is on.

- **MCB (Miniature Circuit Breaker)**

  The MCB is an electromechanical device that switches off the circuit automatically if an abnormality is detected. The MCB easily senses the over current caused by the short circuit. We can use to control the panel and also protect the whole panel.

Figure 3.3.9 SMPS

Figure 3.3.10 MCB

- **SMPS (Switch Mode Power Supplies)**

  A switched-mode power supply (SMPS) is an electronic circuit that converts power using switching devices that are turned on and off at high frequencies, and storage components such as inductors or capacitors to supply power when the switching device is in its non-conduction state. In the project there are 2 SMPS are used one SMPS gave 24V Dc with 2Amp which can control and actuate the proximity sensors, push button and PLC.

The second SMPS gave 12 and 5V Dc with 15Amp current. It can control and actuate the conveyer and Microcontroller and IoT portion used in panel.

- **Relay Card**

  Relays are being used as the interface between the two circuits at same or different potential to achieve. In the project relay card is used to control and actuate the output when signal is given to the input by this we can control Led, conveyor, DCV.

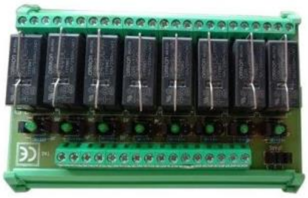

Figure 3.3.11

Figure 3.3.12

- **Proximity Sensor**

  A Proximity sensor is a sensor able to detect the presence of nearby objects without any physical contact. The main use of sensor is detecting the object and control the conveyor withhelp of PLC.

- **Motor**

  A Direct Current (DC) motor is a rotating electrical device that converts direct current, of electrical energy, into mechanical energy. An Inductor (coil) inside the DC motor produces a magnetic field that creates rotary motion as DC voltage is applied to its terminal.

  In the project DC gear motor is used in the conveyor.

Figure 3.3.13 Motor

Figure 3.3.14

- **Buck Convertor**

  Buck convertor is component which can use to control the voltage vary. In the project we can us12V DC to 0 to 12V DC vary Buck convertor for dc motor signal.

- **Wires**

  An electrical wire is a type of conductor, which is a material that conducts electricity. We canuse different wire according to condition.

3.4 Introduction to PLC

A programmable logic controller (PLC) is an industrial computer control system that continuously monitors the state of input devices and makes decisions based upon a custom program to control the state of output devices.

It is an industrial computer control that has been ruggedized and adapted used for the control ofindustrial automation and factories.

PLCs are often used in factories and industrial plants to control motors, pumps, lights, electric supply, etc.

In the major project PLC play an important role to run the whole setup. The main aim of usethe PLC is to control the conveyer setup.

In the project we can use DELTA PLC Model DVP-14SS2. The main specification of PLC is defined as follows:

It is transistor-based PLC.

There are 8 inputs and 6 digital outputs with source and sinking.

It can perform 8K steps in serial programming. It can support serial communication for read or writing the program.

Figure 3.4.1 PLC

3.5 Introduction to WPL Software

In the project DELTA PLC can be used for working. It can be programmed with help of Ladder logic language. For this PLC WPL Soft software can be used for programming.

WPL (World Programming Limited)

WPL Soft is software for PLC (Programmable Logic Controller). It is open-source software in use. When PLC is in operation, uses WPL Soft to monitor the set value or temporarily saved value in Timer (T), Counter (C), and Register (D) and force ON/Off of output contacts. PLC is a control system using electronic operations. It can simulate the ladder program also. There are many functions which can be used in software.

Figure 3.5.1

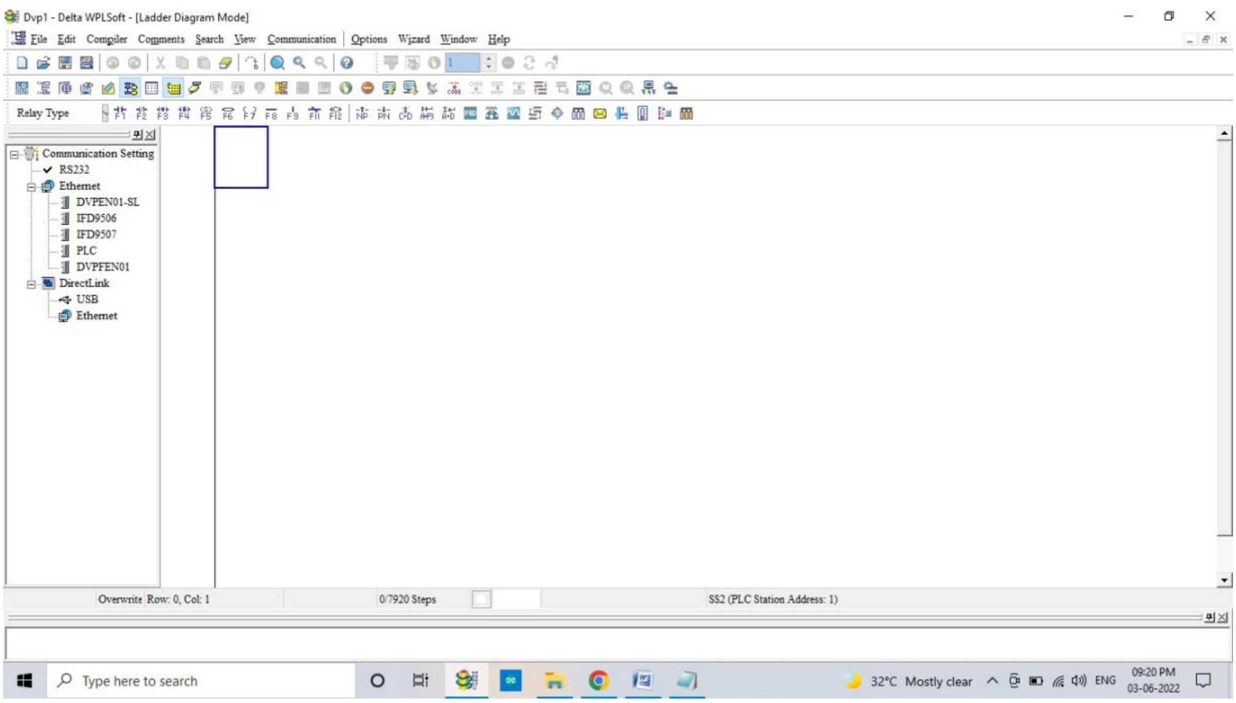

Figure 3.5.2

3.6 Ladder Logic Programming in WPL Software

In the project we can use WPL software for ladder logic programming. According to ourproject requirement the ladder diagram is created which can control the Project and conveyor. The ladder diagram is defined in figure 3.6.1

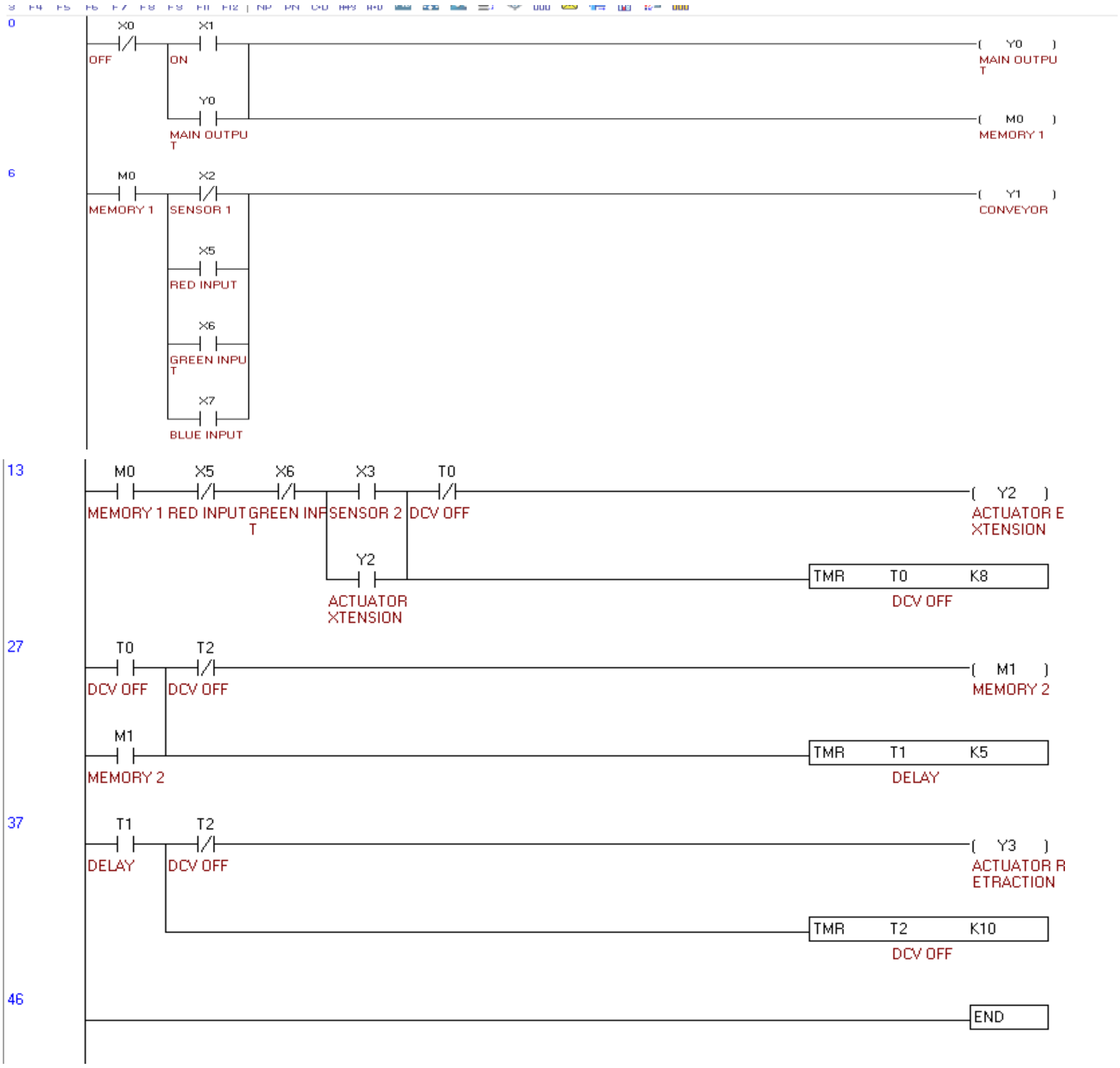

Figure 3.6.1

The above ladder logic is created to control the panel system and also control the conveyor.This ladder logic helps to give the signal through PLC system.

3.7 Detailed Description of PLC Program

In the major project we can use PLC to control the panel and conveyor. First, we can make ladderlogic programming in WPL software and upload the program in the PLC. The detailed description of PLC program is defined as follow:

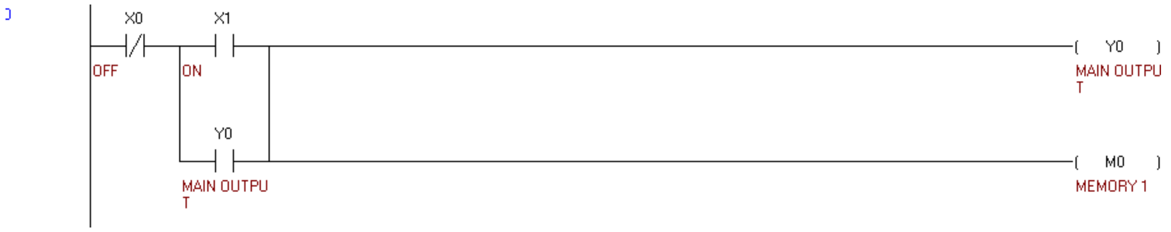

Figure 3.7.1

In the above figure 3.7.1 the latching program is used. In this X0 used as input signal which canbe connected to red push button which can stop the whole working. X1 is used as input signal which can be connected to green button which can run the condition.

Y0 is used as output signal when we gave a signal to X1 then Y0 and M0 become high and gave continuous output.

Figure 3.7.2

In the above figure 3.7.2 we can use M0 NO for signal high. We can use X2 NC for sensor input signal, X5 NO for red colour input signal, X6 NO for green colour input signal, X7 NO for blue colour input signal and Y1 for conveyor output signal.

In this process Y1 is already high. When the sensor gave signal then X2 become NO and Y1 signal become low and stop the conveyor. When the red input signal is given to X5 NO it become NC and Y1 signal become high and start the conveyor. Same condition is applied on green and blue input signal.

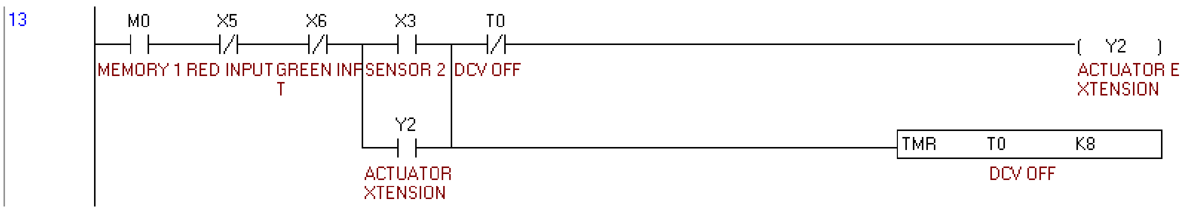

Figure 3.7.3

In the above figure 3.7.3 it can control the actuator signals. So, we can use X3 NO as input signal of sensor. T0 timer NC is used for pulse given to output Y2. Y2 is used as output signal which is connected to DCV relay.

When sensor detect the blue colour object then X3 NO changes to NC and gave signal to Y2 and become high for 1 second with help of timer condition. When the object is sort output signal Y2 become low.

This ladder logic extends the actuator. By this process we can sort the blue colour.

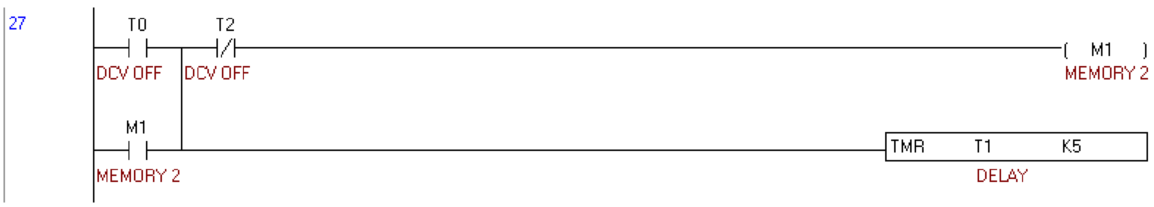

Figure 3.7.4

In the above figure 3.7.4 it can make delay of 500 milliseconds. In this we can use memory M1 NO, Timer T0 NO, T2 NC of timer T2 and Timer T1.

When timer T0 is high then it M1 become also high and timer T1 start it's counting gave the continuous pulse.

Figure 3.7.5

In the above figure 3.7.5 use the T1 N0, Timer T2, T2 NC and output Y3. In this when T1 NO becomes NC then Y3 become high and the timer will give the continuous pulse after 1 second. After that T2 NC changes into NO and reset the timer and Y3 become low.

This Ladder Logic is used to retract the actuator.

Module 4: Wi-Fi Based Microcontroller & IOT Cloud Platform

4.1 Introduction to Internet of Things (IoT) & its implementation

Internet of Things (IoT)

IoT is the networking of physical objects that contain electronics embedded within their architecture in order to communicate and sense interactions amongst each other or with respect to the external environment. In the upcoming years, IoT-based technology will offer advanced levels of services and practically change the way people lead their daily lives. Advancements in medicine, power, gene therapies, agriculture, smart cities, and smart homes are just a very few of the categorical examples where IoT is strongly established.

Over 9 billion 'Things' (physical objects) are currently connected to the Internet, as of now.In the future, this number is expected to rise to a whopping 20 billion.

- **There are Four Main Components Used in IOT:**

1. **Low-Power Embedded Systems:** Less battery consumption, high performance is the inverse factors that play a significant role during the design of electronic systems.

2. **Cloud Computing:** Data collected through IOT devices is massive and this data has to be stored on a reliable storage server. This is where cloud computing comes into play. The data is processed and learned, giving more room for us to discover where things like electrical faults/errors are within the system.

3. **Availability of Big Data:** We know that IOT relies heavily on sensors, especially in real-time. As these electronic devices spread throughout every field, their usage is going to trigger a massive flux of big data.

4. **Networking Connection:** In order to communicate, internet connectivity is a must where each physical object is represented by an IP address. However, there are only a limited number of addresses available according to the IP naming. Due to the growing number of devices, this naming system will not be feasible anymore. Therefore, researchers are looking for another alternative naming system to represent each physical object.

- **There are two ways of building IOT:**

a. Form a separate internet work including only physical objects.

b. Make the Internet ever more expansive, but this requires hard-core technologies such as rigorous cloud computing and rapid big data storage (expensive).
In the near future, IoT will become broader and more complex in terms of scope.

- **IOT Enablers:**

RFIDs: Uses radio waves in order to electronically track the tags attached to each physical object.
Sensors: Devices that are able to detect changes in an environment (ex: motion detectors).
Nanotechnology: As the name suggests, these are extremely small devices with dimensions usually less than a hundred nanometers.
Smart networks: (ex: mesh topology**).**

- **Characteristics of IoT:**

a. Massively scalable and efficient.
b. IP-based addressing will no longer be suitable in the upcoming future.
c. An abundance of physical objects is present that do not use IP, so IoT is made possible.
d. Devices typically consume less power. When not in use, they should be automatically programmed to sleep.
e. A device that is connected to another device right now may not be connected in another instant of time.

- **Intermittent Connectivity** – IoT devices aren't always connected. In order to save bandwidth and battery consumption, devices will be powered off periodically when not in use. Otherwise, connections might turn unreliable and thus prove to be inefficient.
  As a quick note, IoT incorporates trillions of sensors, billions of smart systems, and millionsof applications.

- **Application Domains**: IoT is currently found in four different popular domains:
  1) Manufacturing/Industrial business - 40.2%
  2) Healthcare - 30.3%
  3) Security - 7.7%
  4) Retail - 8.3%

- **Modern Applications:**

1. Smart Grids and energy saving
2. Smart cities
3. Smart homes
4. Healthcare
5. Earthquake detection
6. Radiation detection/hazardous gas detection
7. Smartphone detection
8. Water flow monitoring
9. Traffic monitoring

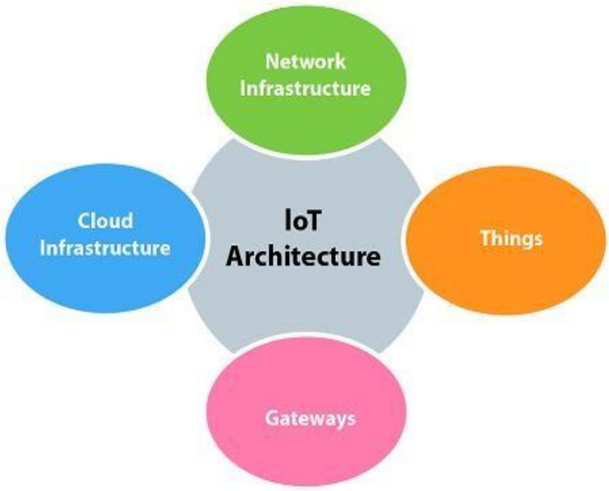

Figure4.1.1 Architecture of IoT

- **Implementation of IoT in This Project: -**A successful implementation requires the integration of IoT components with existing systems.
  In this project we have used IoT to send real time data in think Speak cloud platform. We have used various sensors like TCS3200 colour sensor, to get real time data and send tothing Speak cloud with the help of Node MCU Wi-Fi development module.

a. **TCS3200 Color Sensor;** TCS3200 is the colour-to-frequency programmable converter launched by TAOS Company. As the first RGB color sensor in the industry, it has the single CMOS circuit integrated with configurable silicon photodiode and current frequency converter; also, it integrates three kinds of RGB light filters into the single chip

b. **Node MCU; Node** MCU is an open-source firmware for which open-source prototyping board designs are available. The name "Node MCU" combines "node" and "MCU" (micro-controller unit). The term "Node MCU" strictly speaking refers to the firmware rather than the associated development kits.

Some major step we are taken to get a data and send to cloud platform: -

1. First, we have interfaced the TCS 3200 with Node MCU to get the three different readings from boxes of different colours like Red, Blue, Green.
2. After that with help of Node MCU we send a color data of different colours to ThinkSpeak platform.

4.2 Introduction to Cloud platform and Thingspeak

A cloud platform refers to the operating system and hardware of a server in an Internet-based data center. It allows software and hardware products to co-exist remotely and at scale. So, howdo cloud platforms work? Enterprises rent access to compute services, such as servers, databases, storage, analytics, networking, software, and intelligence. Therefore, the enterprises don't have to set up and own data centers or computing infrastructure.They simply pay for what they use.
There are several types of cloud platforms. Not a single one works for everyone. There areseveral models, types, and services available to help meet the varying needs of users. They include:

- **Public Cloud**: Public cloud platforms are third-party providers that deliver computing resources over the Internet. Examples include Amazon Web Services (AWS), Google Cloud Platform, Alibaba, Microsoft Azure, and IBM Bluemix.
- **Private Cloud**: A private cloud platform is exclusive to a single organization. It's usually in an on-site data center or hosted by a third-party service provider.
- **Hybrid Cloud**: This is a combination of public and private cloud platforms. Data and applications move seamlessly between the two. This gives the organization greater flexibility and helps optimize infrastructure, security, and compliance.

In our project we have used THINGSPEAK cloud platform so brief description of ThingSpeak is given below: -
ThingSpeak is an IoT analytics platform service that allows you to aggregate, visualize and analyze live data streams in the cloud. ThingSpeak provides instant visualizations of dataposted by your devices to ThingSpeak.

Getting Started

Step 1 = Open ThingSpeak and click on the 'Get Started Now' button on the centerof the page and you will be redirected to the sign-up page (you will reach the same page when you click the 'Sign Up' button on the extreme right). Fill out the required details and click on the 'Create Account' button.

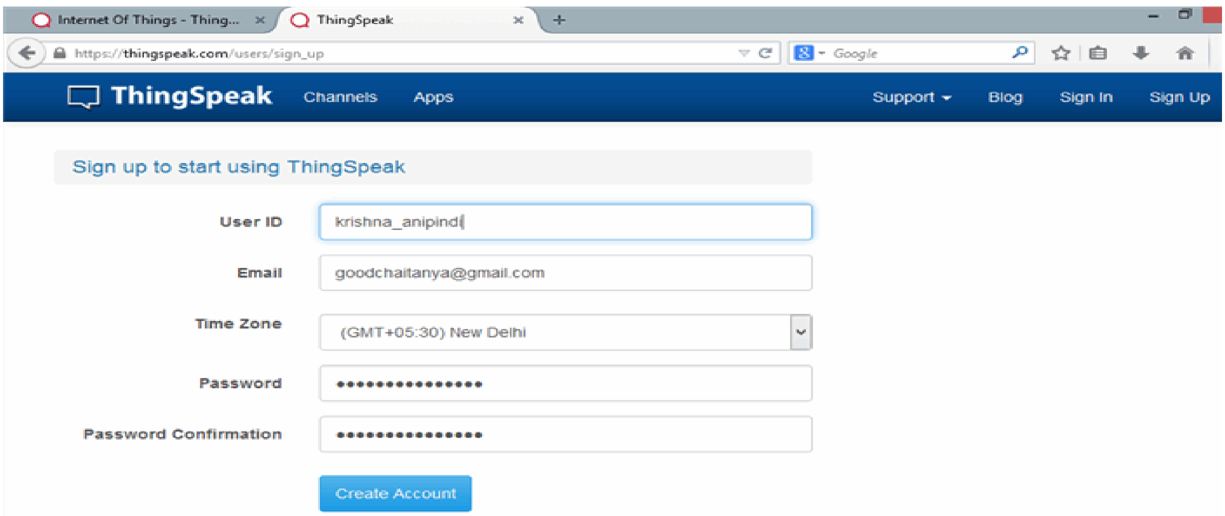

Figure4.2.1 Account step process.

Step 2 = Now you should see a page with a confirmation that the account was successfully created. The confirmation message disappears after a few seconds and the final page shouldlook as in the below screen:

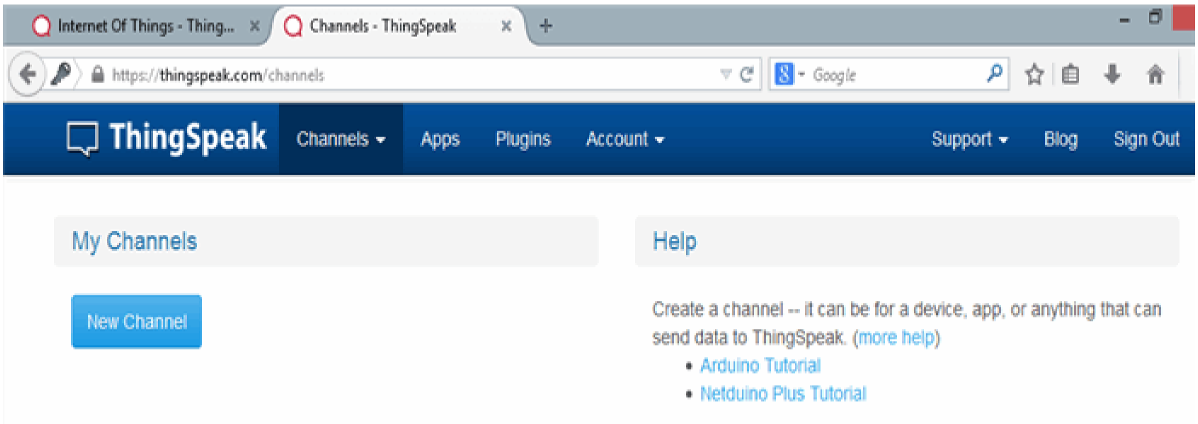

Figure4.2.2 Successfully created channel.

Step 3 = Go ahead and click on 'New Channel'. You should see a page like the below:

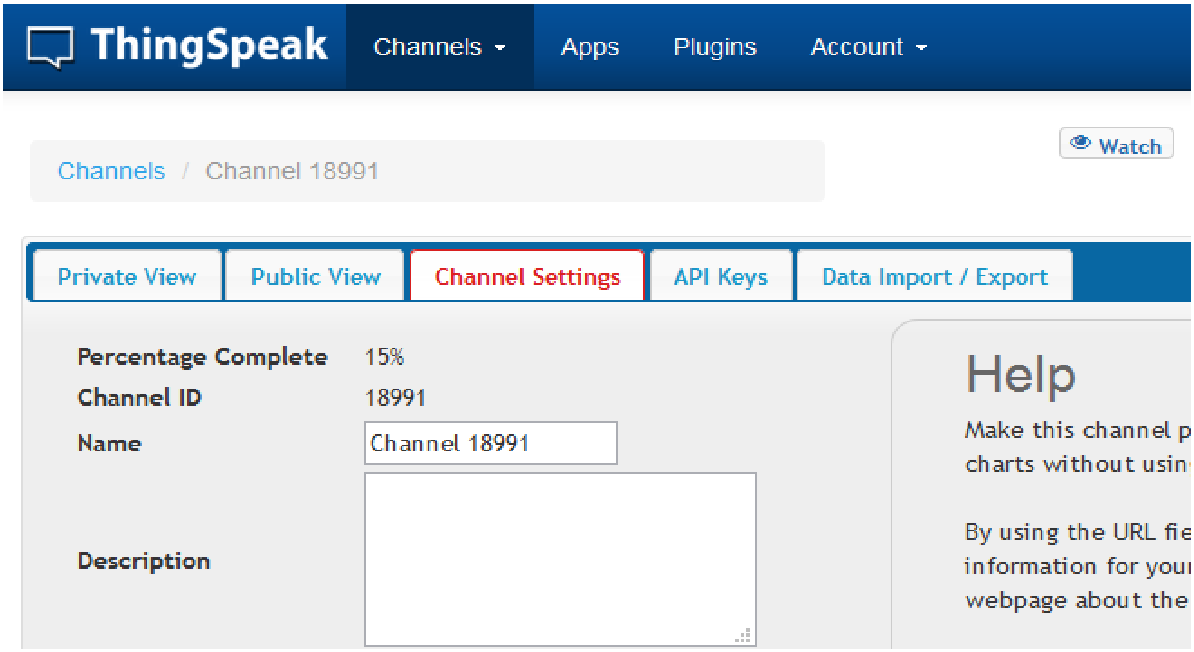

Figure4.2.3 Channel Settings

You can change the name to fit your need and you can add a description corresponding to the channel. You can add any other useful description into the metadata field. In the same page, you should see the fields for Latitude, Longitude and Elevation. Also, when you scroll down you should see a check box that says 'Make Public?'. Let us consider the significance of the various fields and the tabs:

- Latitude, longitude and elevation - These fields correspond to the location of a 'thing' and are especially significant for moving things.
- Make Public? - If the channel is made public, anyone can view the channel's data feed and the corresponding charts. If this check box is not checked, the channel is private, which means for every read or write operation, the user has to pass a corresponding API key.
- URL - This can be the URL of your blog or website and if specified, will appear on the public view of the channel.
- Video ID - This is the ID corresponding to your YouTube or Video ID. If specified, the video appears on the public view of the channel.

- Fields 1 to 8 - These are the fields which correspond to the data sent by a sensor or a 'thing'. A field has to be added before it can be used to store data. By default, Field 1 is added. In case you try posting to fields that you have not added, your request will still be successful, but you will not be able to see the field in the charts and the corresponding data. You can click on the small box before the 'add field' text corresponding to each field to add it. Once you click the 'add field' box, a default label name appears in the text box corresponding to each field and the 'add field' text changes to 'remove field'. You can edit the field text that appears by default when a field is added to make more sense. For example, in the below screen, I have modified the text for Field 2 to 'Sensor Input'. To remove a field which is added, just check on the 'remove field' box. Once you click this, the text 'remove field' changes back to 'add field' and the corresponding field text is cleared.

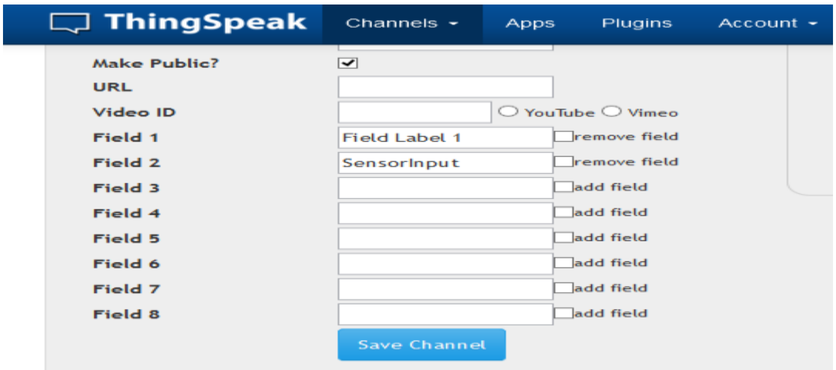

Figure 4.2.4 Channel selection.

Step 4 = Once you have edited the fields, click on 'Save Channel' button. You should now seea page like the below in which the 'Private View' tab is defaulted.

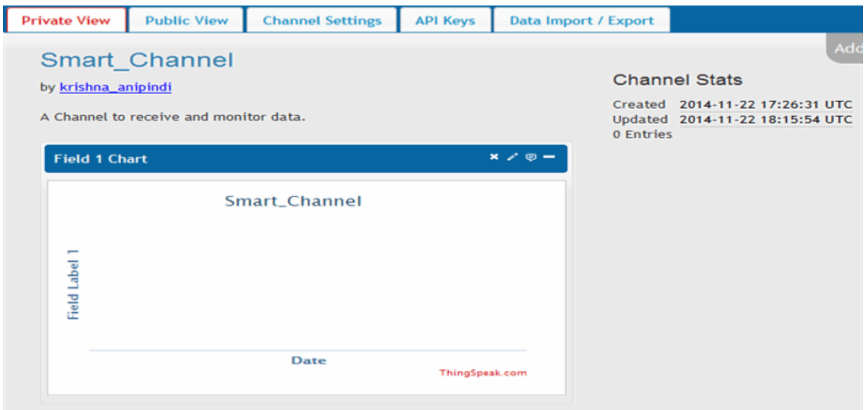

Figure 4.2.5 Channel View

The Private View shows a chart corresponding to each of the fields that we have added. Now click on the 'Public View' tab. This should look exactly similar to what we see in the'Private View' tab since our channel is public. In case your channel is not public ('make public' check box not checked in the 'channel settings' tab), the public view tab shows amessage that 'This channel is not public'.

Step 5 = Now click on the 'API Keys' tab. You should see a screen similar to the below. Thewrite API key is used for sending data to the channel and the read API key(s) is used to read the channel data. When we create a channel, by default, a write API key is generated. We generate read API keys by clicking the 'Generate New Read API Key' button under thistab. You can also add a note corresponding to each of the read API keys.

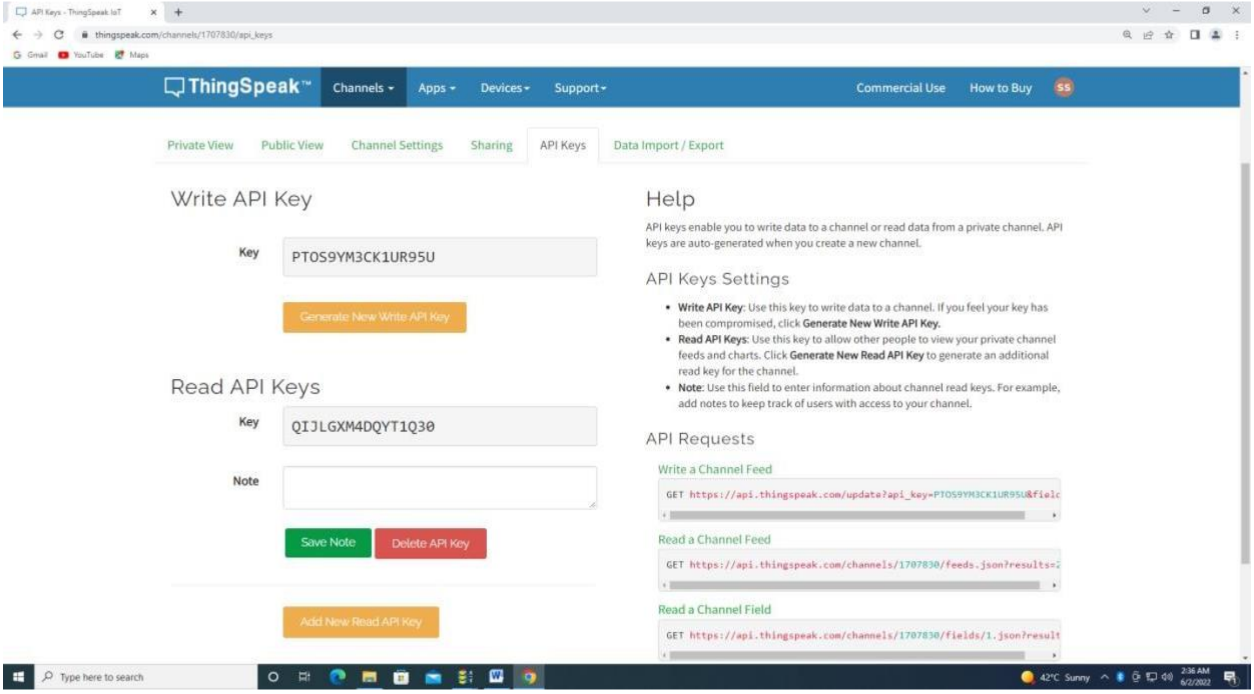

Figure 4.2.6 API key.

4.3 Steps to Design the Dashboard for Real Time Data Monitoring.

- Open the ThingSpeak official website and log in your channel.
- Go to channel setting. And fill the detail like,

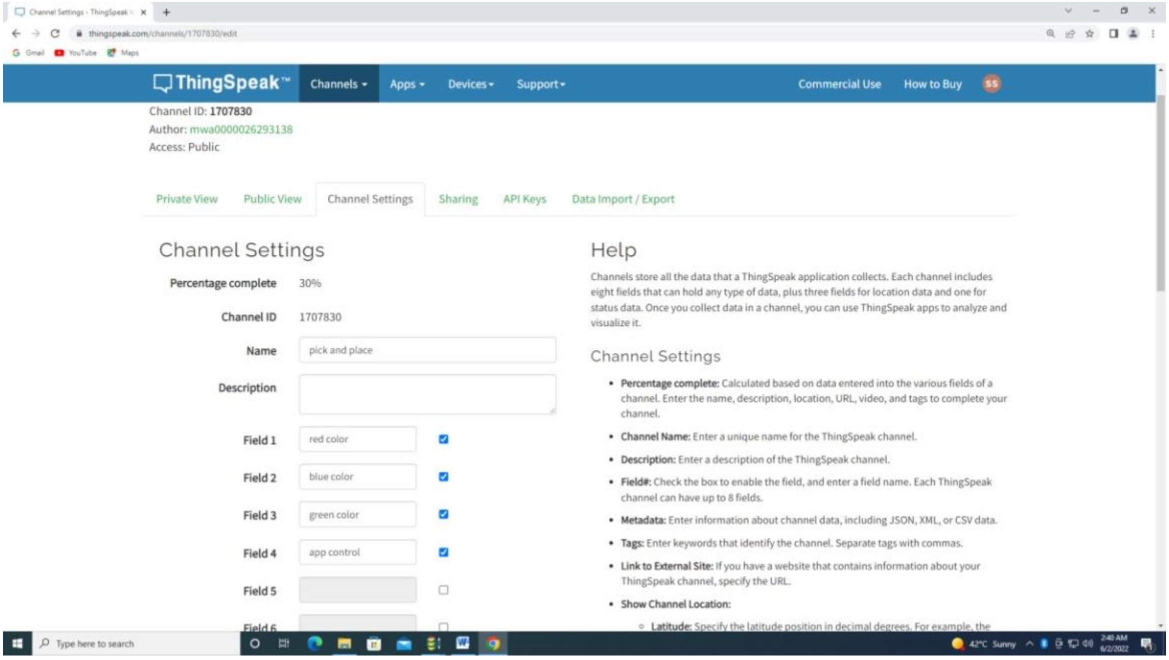

Figure 4.3.1. Dashboard setup.

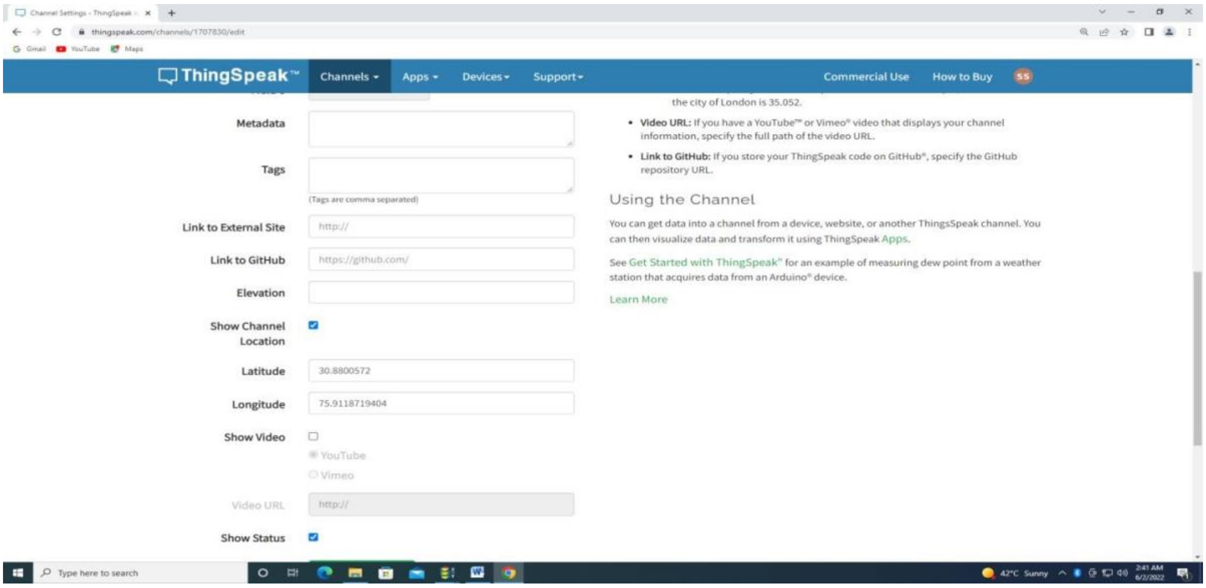

Figure 4.3.2. Dashboard setup.

- Click the “OK” Button.
- Go to the ADD Widgets
- Add numeric display.

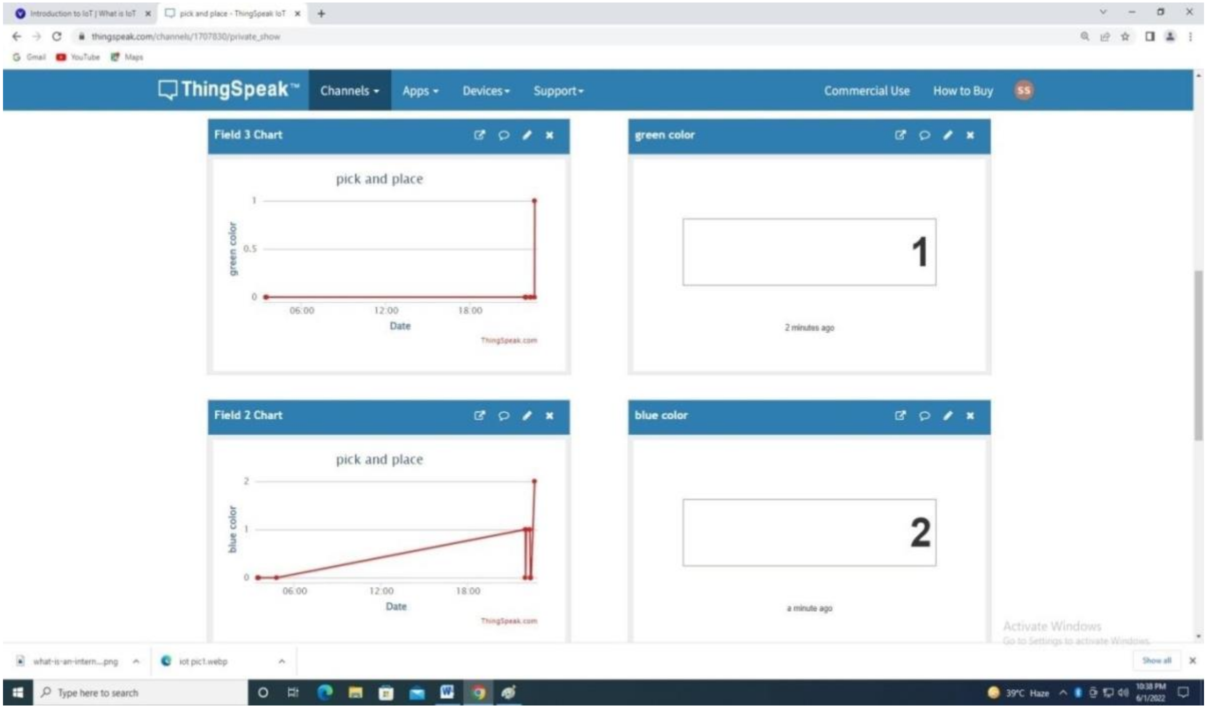

Figure 4.3.3 Real time Data Monitoring.

4.4 Bill of Material

The component use for microcontroller is detailed in BOM in table 4.1

Sr. no	Material name	Specification of material	Unit	Quantity
1	Microcontroller	Node MCU 8266 Wi-Fi based module used forIoT	No.	01
2	Color sensor	TCS3200. Operated voltage 2.7 to 5.5V DC,High resolution	No.	01
3	Servo motor	MG995. 5V DC operated, Metal gear and hightorque.	No.	02
4	Servo motor 9G	5v operated, Plastic gear, less torque	No.	1
5	Relay module	4 channel, 5V DC operated	No.	01
6	Relay module	2 channel 5V DC operated	No.	01
7	General PCB Board	Size 4*6inch	No.	02
8	Microcontroller	Arduino uno ATMega328p, 14 Digital or 6analog pins.	No.	01
9	Bluetooth	HC-05, 3.3 to 5V DC operated.	No.	01
10	Ribbon wire	10 channels	No.	01
11	Header	Male or Female	Strip	03
12	Connecter	Male or Female	No.	10
13	Gripper	Acrylic material	No.	01
14	SMPS	220V operated, Output = 12, 5, 3.3V DC15amp.	No.	01

Table 4.4.1 Bill of material

4.5 Circuit Diagram

First, we can use Auto cad electrical for make the panel design. After the completing the design further electrical panel or connection of component will make. There are two designs whichcan be made in Auto cad as defined in figure:

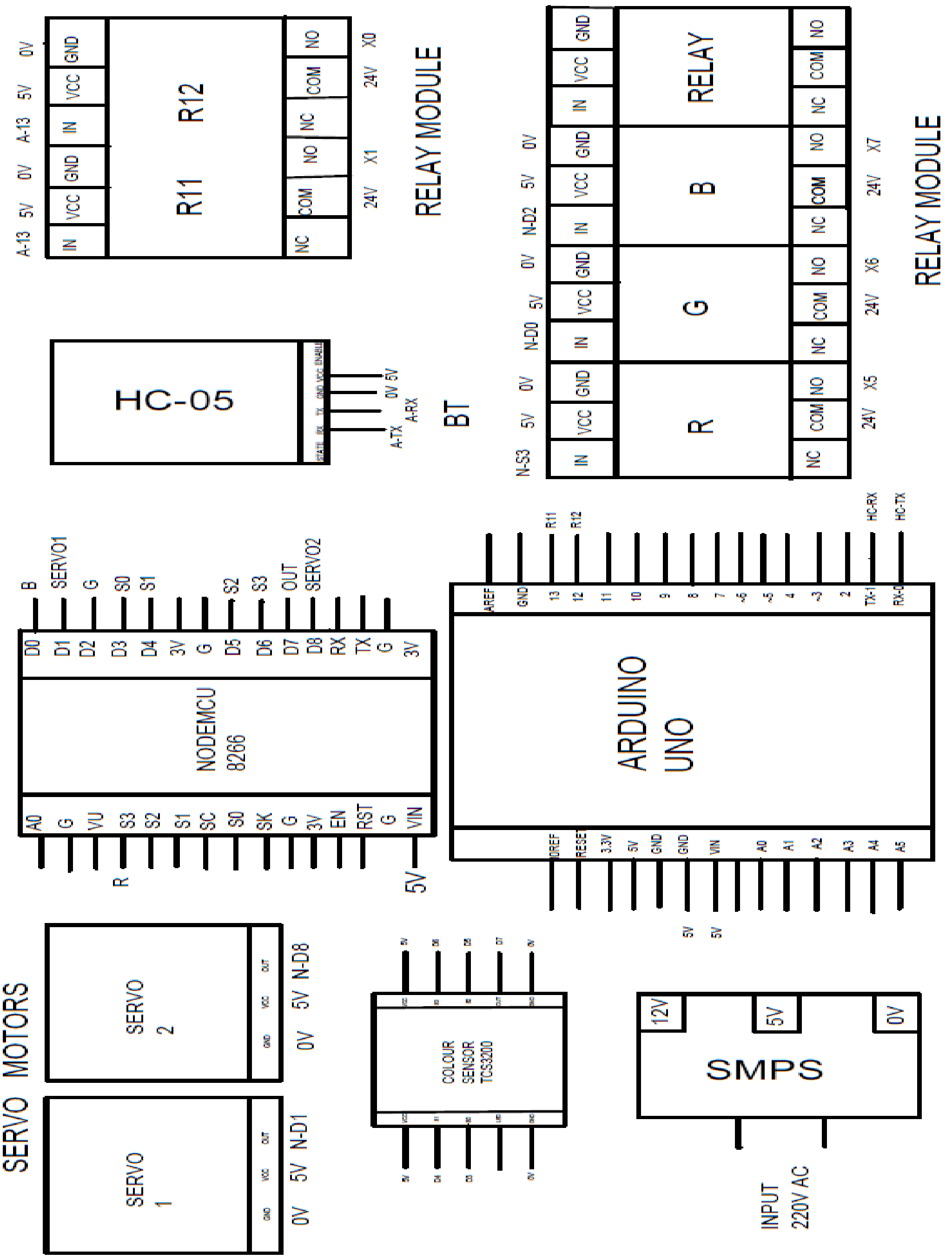

Figure 4.5.1 Circuit diagram

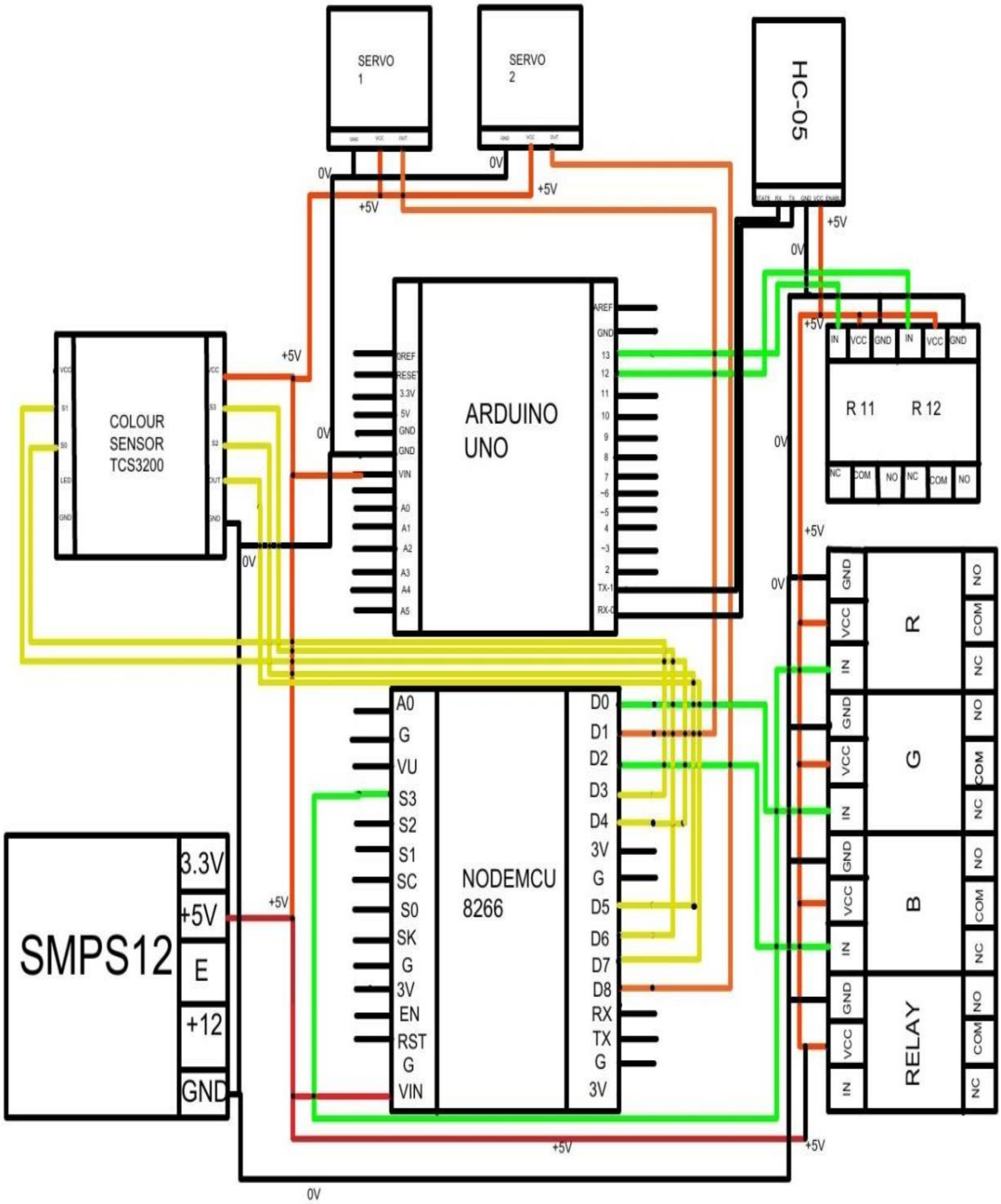

Figure 4.5.2 Circuit diagram

4.6 Introduction to Different Electronics Components

- **<u>Node MCU:</u> -** Node MCU is an open source IoT platform. It includes firmware which runs on the ESP8266 Wi-Fi SoC from Espressif Systems, and hardware, which is based on the ESP-12 module. The term "Node MCU" by default refers to the firm ware rather than the dev kits. The firmware uses the Lua scripting language.

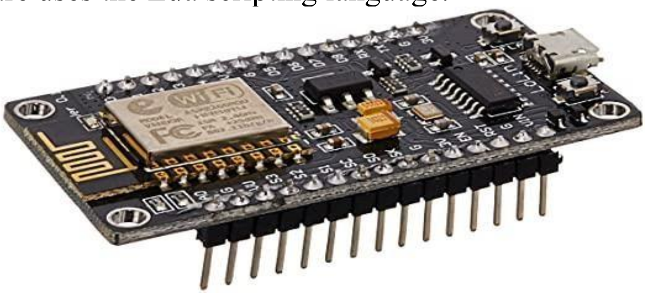

Figure 4.6.1 Node MCU.

How to Write Codes for Node MCU?

After setting up ESP8266 with Node-MCU firmware, let's see the IDE (Integrated Development Environment) required for the development of Node MCU. Node MCU with ESPlorerIDE Lua scripts are generally used to code the Node MCU. Lua is an open-source, lightweight, embeddable scripting language built on top of C programming language. For more information about how to write Lua script for NodeMCU refer to Getting started with NodeMCU using ESPlorerIDE Node MCU with Arduino IDE Here is another way of developing Node MCU with a well-known IDE i.e. Arduino IDE. We can also develop applications on Node MCU using the Arduino development environment. This makes it easy forArduino developers than learning a new language and IDE for Node MCU. For more information about how to write Arduino sketch for Node MCU refer to Getting started with Node MCU using Arduino IDE

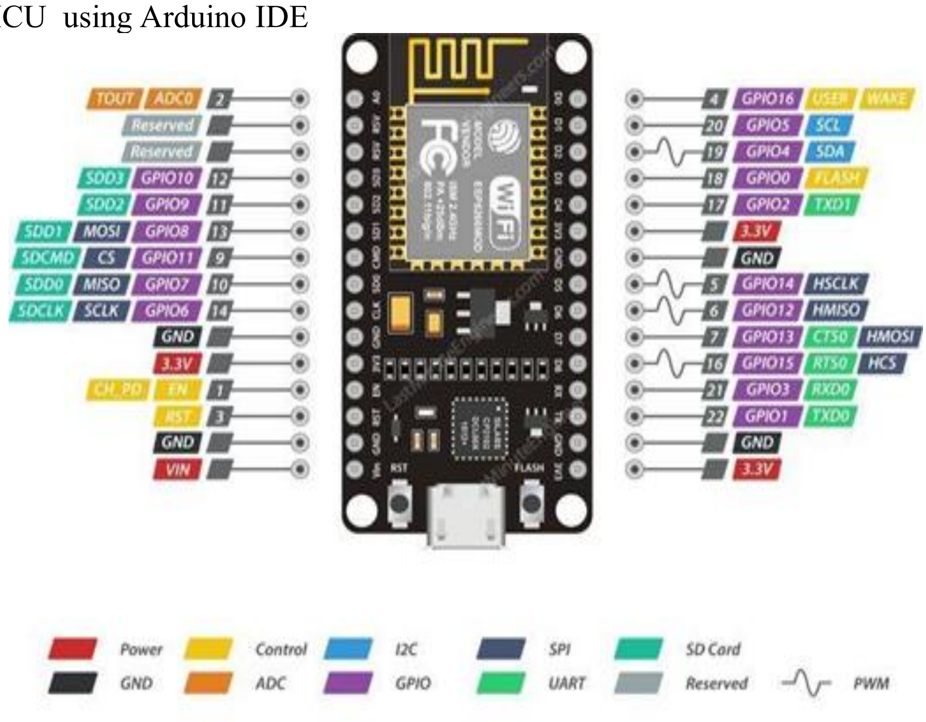

Figure 4.6.2 pin detail of Node MCU.

- **TCS3200 color Sensor**

TCS3200 is the colormap-to-frequency programmable converter launched by TAOS Company.As the first RGB color sensor in the industry, it has the single CMOS circuit integrated with configurable silicon photo diode and current frequency converter; also, it integrates three kinds of RGB light filters into the single chip.

1. Operating Voltage: 3V-5V DC
2. High-resolution conversion of light intensity to frequency.
3. Chip pins all has drawn for standard 100.
4. Put the needle (2.54 mm).
5. Mil-convenient for bitmap board.
6. Programmable color and full-scale output frequency.
7. Communicate directly with a microcontroller.
8. Low-profile surface mount package.

TCS3200 has an array of photodiodes with 4 different filters. A photodiode is simply asemiconductor device that converts light into current. The sensor has:
16 photodiodes with red filter – sensitive to red wavelength
16 photodiodes with green filter – sensitive to green
wavelength16 photodiodes with blue filter – sensitive to
blue wavelength 16 photodiodes without filter

Figure 4.6.3 color Sensor

Pin Name	I/O	Description
GND (4)		Power supply ground
OE (3)	I	Enable for output frequency (active low)
OUT (6)	O	Output frequency
S0, S1 （1， 2）	I	Output frequency scaling selection inputs
S2, S3 （7， 8）	I	Photodiode type selection inputs
VDD （5）		Voltage supply

Table 4.6.1 Pin detail of colour sensor.

Frequency Scaling

Pins S0 and S1 are used for scaling the output frequency. It can be scaled to the following preset values: 100%, 20% or 2%. Scaling the output frequency is useful to optimize the sensor readings for various frequency counters or microcontrollers. Take a look at the table below:

Output frequency scaling	S0	S1
Power down	L	L
2%	L	H
20%	H	L
100%	H	H

Table4.6.2Pin details of color sensor

For the Arduino, it is common to use a frequency scaling of 20%. So, you set the S0 pin to HIGH and the S1 pin to LOW.

- **Servo Motor=**A servo motor is an electromechanical device that produces torque and velocity based on the supplied current and voltage. A servo motor works as part of a closed loop system providing torque and velocity as commanded from a servo controller utilizing afeedback device to close the loop.**MG995** servo is a simple, commonly used standard servo for your mechanical needs such as robotic head, robotic arm. It comes with a standard 3-pinpower and control cable for easy using and metal gears for high torque. A Me RJ25 Adapteralso helps you to connect the servo with Me Baseboard or Make block Orion easily.

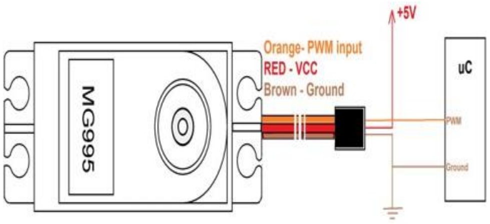

Figure 4.6.4 Servo Motor with pin diagram.

- **Relay Module: -**A power relay module is an electrical switch that is operated by an electro magnet. The electromagnet is activated by a separate low-power signal from a micro controller. When activated, the electromagnet pulls to either open or close an electrical circuit. You can use a 5V relay to switch the 120-240V current and use the Arduino to control the relay. * A relay basically allows a relatively low voltage to easily control higher power circuits.

Figure 4.6.5 Relay Module

- **PCB: -**As its name suggests, general purpose PCBs are widely used to embed circuits randomly for running of hardware. Its layer is coated with copper and allows proper soldering without any short circuit. General purpose board, connections are not built so connections are to be created

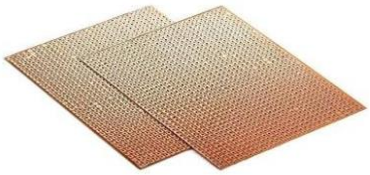

Figure 4.6.6 PCB

- **Arduino UNO**

  Arduino UNO is a low-cost, flexible, and easy-to-use programmable open-source microcontroller board that can be integrated into a variety of electronic projects. This board canbe interfaced with other Arduino boards, Arduino shields, Raspberry Pi boards and can control relays, LEDs, servos, and motors as an output.

  - Microcontroller: MicrochipATmega328P
  - Operating Voltage: 5 Volts
  - Input Voltage: 7 to 20 Volts
  - Digital I/O Pins: 14 (of which 6 can provide PWM output)
  - PWM Pins: 6 (Pin # 3, 5, 6, 9, 10 and 11)[9]
  - UART: 1
  - I2C: 1
  - SPI: 1
  - Analog Input Pins: 6
  - DC Current per I/O Pin: 20 mA
  - DC Current for 3.3V Pin: 50 mA
  - Flash Memory: 32 KB of which 0.5 KB used by boot loader
  - SRAM: 2 KB
  - EEPROM: 1 KB
  - Clock Speed: 16 MHz
  - Length: 68.6 mm
  - Width: 53.4 mm
  - Weight: 25 g
  - ICSP Header: Yes
  - Power Sources: DC Power Jack & USB Port

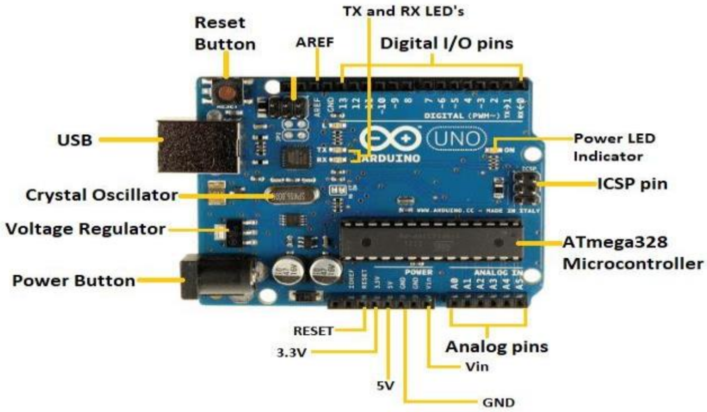

Figure 4.6.6 Arduino Uno

- **Bluetooth HC--05.**

a. It is used for many applications like wireless headset, game controllers, wireless mouse,wireless keyboard and many more consumer applications.

b. It has a range up to <100m which depends upon transmitter and receiver, atmosphere,geographic & urban conditions.

c. It is IEEE 802.15.1 standardized protocol, through which one can build wireless Personal Area Network (PAN). It uses frequency-hopping spread spectrum (FHSS) radio technologyto send data over air.

d. It uses serial communication to communicate with devices. It communicates withmicrocontroller using serial port (USART)

e. Pin Description

Figure 4.6.7 HC-05 BlueTooth

- Bluetooth serial modules allow all serial enabled devices to communicate with eachother using Bluetooth.
- It has 6 pins,
- 1. Key/EN: It is used to bring Bluetooth module in AT commands mode. If Key/EN pinis set to high, then this module will work in command mode. Otherwise by default it is in data mode. The default baud rate of HC-05 in command mode is 38400bps and 9600 in data mode.
- HC-05 module has two modes,
  1. Data mode: Exchange of data between devices.
  2. Command mode: It uses AT commands which are used to change setting of HC-05. To send these commands to module serial (USART) port is used.
- 2. VCC: Connect 5 V or 3.3 V to this Pin.
- 3. GND: Ground Pin of module.
- 4. TXD: Transmit Serial data (wirelessly received data by Bluetooth moduletransmitted out serially on TXD pin)
- 5. RXD: Receive data serially (received data will be transmitted wirelessly byBluetooth module).
- 6. State: It tells whether a module is connected or not.

- Ribbon cabling is used for data transmission and communications. It is often chosen for computer cable applications, where it is used as internal wiring for hard drives, CD drives, and more. Ribbon cables are also commonly used as internal wiring for other electronics and appliances.

Figure 4.6.8 Ribbon Wire

- **Pin Header**
  Pin header are stiff metallic connectors that are soldered to a circuit board and stick up to receive a connection from a female socket. While pin headers (often called PH, or headers) are male by definition, female equivalents are also quite common, and we refer to them as female headers (FH) or header connectors.

- **Gripper**
  In the simplest terms, grippers are devices that enable robots to pick up and hold objects. When combined with a collaborative robot (or 'robot') arm, grippers enable manufacturers to automate key processes, such as inspection, assembly, pick & place and machine tending.
  This servo gripper is used for the pick and place robot mechanism. This product is generally used by students for their projects. It is also used for industrial uses. This product is made of acrylic sheets.

Figure 4.6.9 Servo Gripper

- **MIT app Inventor**
  MIT App Inventor is an intuitive, visual programming environment that allows everyone even children to build fully functional apps for smartphones and tablets. Those new to MIT App Inventor can have a simple first app up and running in less than 30 minutes.

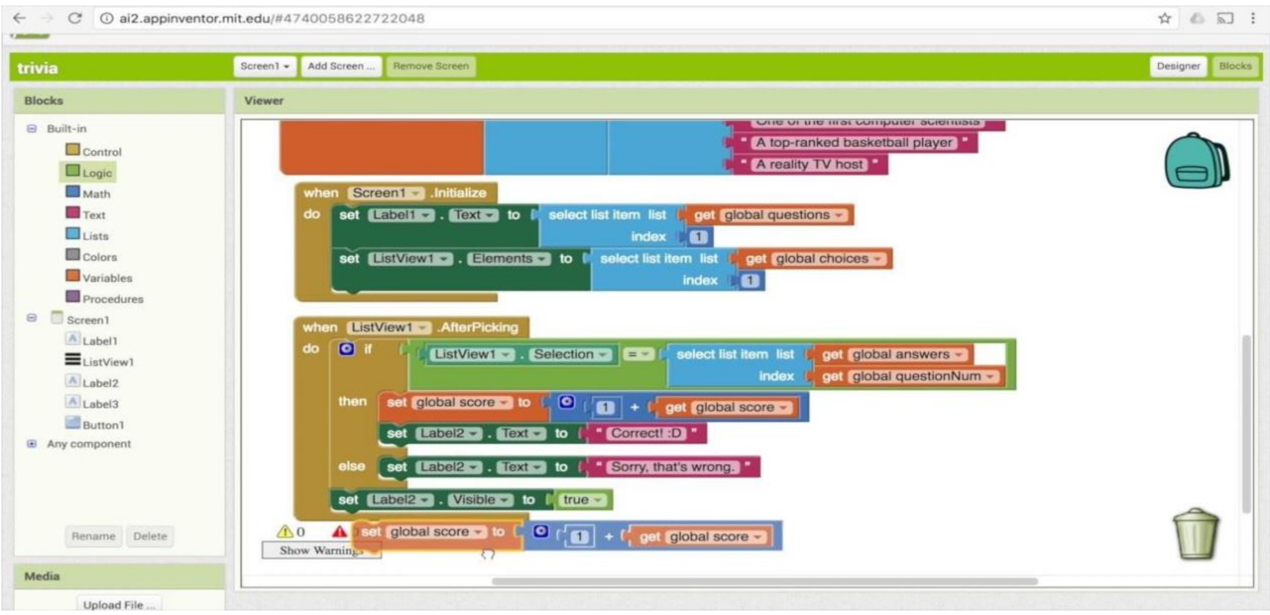

Figure 4.6.10

4.7 Embedded C Program

Here the programming is written in table which can be used in Microcontroller.

#include<ESP8266WiFi.h>	delay(3000);
#include <ThingSpeak.h>	digitalWrite(relay1,LOW);
#include <Servo.h>	flag=1;
WiFiClient client;	myservo.write(85);
long mychannelNumber=1707830;	delay(1000);
const char myWriteAPIKey[]="PTOS9YM3CK1UR95U";	myservo1.write(160);
Servo myservo;	delay(300);
Servo myservo1;	digitalWrite(relay1,LOW);
constint fieldnumber1=1;	flag=1;
constint fieldnumber2=2;	myservo.write(85);
constint fieldnumber3=3;	delay(1000);
constint s0=D3;	myservo1.write(160);
constint s1=D4;	delay(300);
constint s2=D5;	myservo1.write(105);
constint s3=D6;	delay(1000);
constint out=D7;	myservo.write(180);
int flag=0;	delay(1000);
int flag1=0;	myservo1.write(160);
int flag2=0;	delay(1000);
int red=0;	myservo.write(85);
int green=0;	delay(1000);
int blue=0;	else if(red>50 && flag==1)
int redcolor=0;	{
intgreencolor=0;	flag=0;
intbluecolor=0;	}
intyellowcolor=0;	else
int color=0;	{
int relay1=10;	Serial.print("redcolor=");
int relay2=D2;	Serial.println(redcolor);
int relay3=D0;	}

Table 4.7.1 Embedded C program.

```
void setup()
{
  myservo.write(85);
   myservo1.write(160);
Serial.begin(9600);
pinMode(s0,OUTPUT);
pinMode(s1,OUTPUT);
pinMode(s2,OUTPUT);
pinMode(s3,OUTPUT);
pinMode(10,OUTPUT);
pinMode(D2,OUTPUT);
pinMode(D0,OUTPUT);
myservo.attach(15);
myservo1.attach(5);
pinMode(out,INPUT);
digitalWrite(s0,HIGH);
digitalWrite(s1,HIGH);
digitalWrite(10,LOW);
digitalWrite(D2,LOW);
digitalWrite(D0,LOW);
WiFi.begin("realme 8","12345678");
while(WiFi.status()!=WL_CONNECTED)
  {
    delay(200);
Serial.print(". .");
 }
 Serial.println();
Serial.println("Node MCU  is
connected!!");
Serial.println(WiFi.localIP());
ThingSpeak.begin(client);
}
void loop() {
 digitalWrite(s2,LOW);
```

```
ThingSpeak.writeField
(mychannelNumber, 2,bluecolor ,
myWriteAPIKey);
  if(blue<30 && flag1==0)
{
bluecolor++;
   Serial.println( bluecolor,DEC);
 Serial.println(blue);
 digitalWrite(relay2,HIGH);
 flag1=1;
 delay(1000);
digitalWrite(relay2,LOW);

}
else if(blue>30 &&  flag1==1)
{
  flag1=0;
}
  else

  {
Serial.print(" bluecolor=");
  Serial.println( bluecolor);
}
ThingSpeak.writeField
(mychannelNumber, 3,greencolor ,
myWriteAPIKey);
if(green<52   &&   red>79   &&
flag2==0)
{
greencolor++;
   Serial.println( greencolor,DEC);
```

```
 digitalWrite(s3,LOW);
 red=pulseIn(out,digitalRead(out)==HIGH?LOW:HIGH);
 digitalWrite(s3,HIGH);

blue=pulseIn(out,digitalRead(out)==HIGH?LOW:HIGH);
 digitalWrite(s2,HIGH);

green=pulseIn(out,digitalRead(out)==HIGH?LOW:HIGH
);

Serial.print("R Intensity");
Serial.println(red,DEC);
Serial.print("G Intensity");
Serial.println(green,DEC);
Serial.print("B Intensity");
Serial.println(blue,DEC);
delay(1000);
ThingSpeak.writeField (mychannelNumber, 1,redcolor ,
myWriteAPIKey);
if(red<50&& flag==0)
{
   redcolor++;
  Serial.println(redcolor,DEC);
  Serial.println(red);
  digitalWrite(relay1,HIGH);
  delay(3000);
  digitalWrite(relay1,LOW);
  flag=1;
  myservo.write(85);
 delay(1000);
 myservo1.write(160);
 delay(300);
}
```

```
  Serial.println(green);
 digitalWrite(relay3,HIGH);
 delay(3000);
 digitalWrite(relay3,LOW);
 flag2=1;
 myservo.write(85);
 delay(1000);
 myservo1.write(160);
 delay(300);
 myservo1.write(105);
 delay(1000);
 myservo.write(0);
 delay(1000);
 myservo1.write(160);
 delay(1000);

 myservo.write(85);
 delay(1000);

  digitalWrite(relay3,LOW);

}
else if(green<52 && red>79 &&
blue<46 && flag2==1)
{
  flag2=0;
}
  else

   {
  Serial.print("greencolor=");
  Serial.println( greencolor);
}
}
```

- **Mobile App Controlling Program:**

Table 4.7.1

```
int relay1=13;
int relay2=12;
int data;
void setup()
{
 pinMode(relay1,OUTPUT);
 pinMode(relay2,,OUTPUT);
 Serial.begin(9600);
}

void loop()
{
if(Serial.available()>0)
 {
 data=Serial.read();
 if(data==1)
 {
digitalWrite(relay1,LOW);
digitalWrite(relay2,HIGH);
delay(1000);
digitalWrite(relay1,LOW);
digitalWrite(relay2,LOW);
 }
 else if(data==2)
 {
  digitalWrite(relay1,HIGH);
digitalWrite(relay2,LOW);
delay(1000);
digitalWrite(relay2,LOW);
digitalWrite(relay1,LOW);
}
```

4.8Description of Embedded C Program With Cloud Real time Data's Screenshots

- In this program first we interface the TCS3200 color sensor with Node MCU.
- After that we can receive three different colors Output reading like (Red, Green and Blue) with help of TCS3200.
- We have declared three variable red color, blue color, green color initializes with zero.
- Store the reading of the color in this declare variable.
- According to the reading make a condition on it or find a specific color.
- With the reading of color sensor, we can ON a relay.
- When the relay gets ON then we can provide some delay because of the product should be comes to close to the gripper.
- When product comes to gripper the gripper picks up the product and according to thecolor of the product and palace to on its position.
- Second app controlling program we generate an app with the help of MIT app inventor.
- In program First we declare Three variable relay1, relay2 and data.
- Store the value of Serial.available function in data variable then make a logic according the give number and create a single pulse with the help of relay.

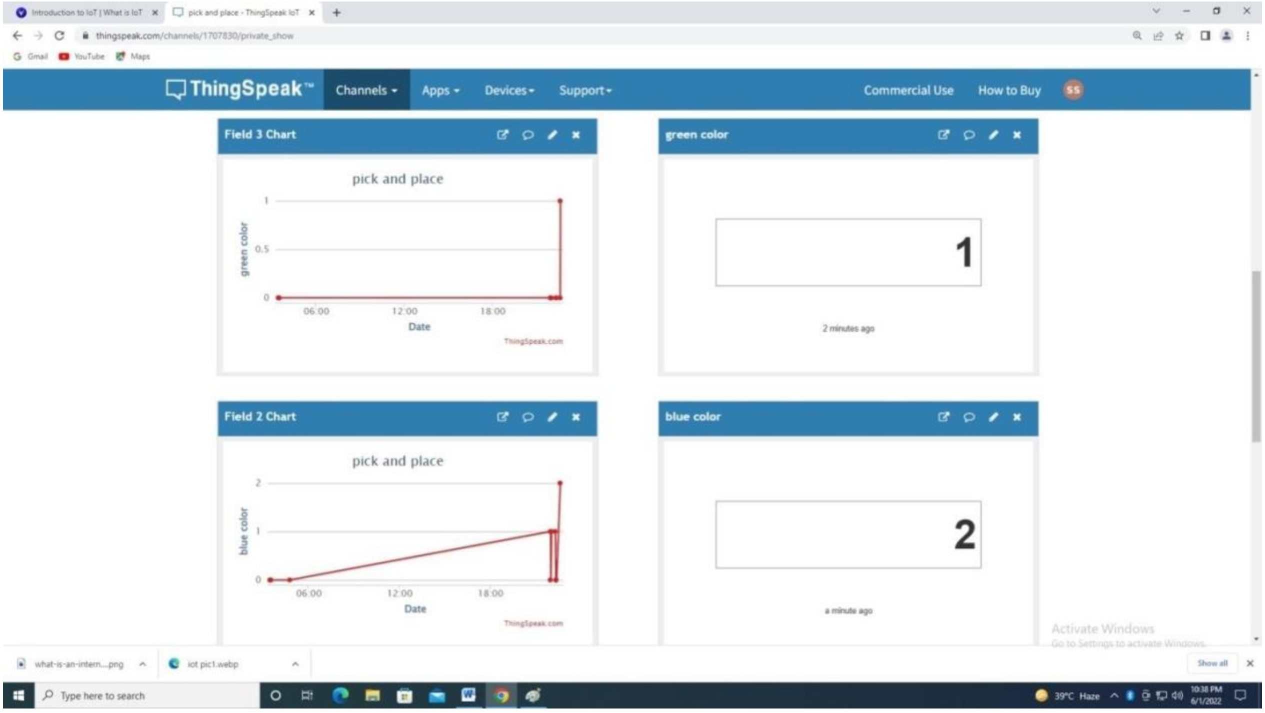

Figure 4.8 Real Time data

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3. https://appinventor.mit.edu/
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